

User Experience

2023/2024

FCUL – Departamento de Informática

Carlos Duarte

Agenda

- UX Design
 - Accessibility
 - Information architecture

DESIGN FOR ACCESSIBILITY

Topics

- Visual Design
- Information Design
- Navigation Design
- Interaction Design
- Images and Graphics
- Multimedia and Animations

Visual Design – Colour

- Sufficient contrast ratio is essential for people with disabilities to perceive, distinguish, and understand content
- Some people rely on colour differences to understand and distinguish content
- Some people perceive colour in different ways, and that some may not perceive colour at all
 - Supplement information presented using colour, using shapes and icons

Visual Design – Colour – Examples

White: #FFFFFF Blue: #6FA3D9 Contrast ratio: 2.65:1

Low contrast color combinations make reading difficult

Low contrast color combinations make reading difficult

White: #FFFFFF Blue: #224367 Contrast ratio: 10.16:1

High contrast color combinations make reading easier

High contrast color combinations make reading easier

Visual Design – Colour – Examples

Today's Forecast

Tomorrow's Forecast

Visual Design – Styles

- People with disabilities rely on distinguishable and customizable style properties, such as font types, sizes, spacing, and text alignment to support readability

Visual Design – Styles – Examples

Welcome to Our Website

This is a sample web page with various styled elements.

[Click here to visit our homepage](#)

Click me

Username:

Password:

Please enter your login credentials.

Thank you for visiting our website. [Contact us](#) if you have any questions.

Visual Design – Styles – Examples

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin vel arcu non urna varius congue sed id libero. Fusce tincidunt interdum malesuada. Integer scelerisque, justo vel vulputate iaculis, tortor erat vestibulum nulla, vel vehicula tellus arcu eget orci.

Nullam eget nisi eu nunc ultricies malesuada. Pellentesque dignissim in est ac varius. Suspendisse potenti. Praesent et fringilla purus, et convallis massa.

Vestibulum et finibus massa. Donec id fermentum velit. Curabitur at ipsum sit amet risus suscipit scelerisque. In ultricies, odio at volutpat.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin vel arcu non urna varius congue sed id libero. Fusce tincidunt interdum malesuada. Integer scelerisque, justo vel vulputate iaculis, tortor erat vestibulum nulla, vel vehicula tellus arcu eget orci.

Nullam eget nisi eu nunc ultricies malesuada. Pellentesque dignissim in est ac varius. Suspendisse potenti. Praesent et fringilla purus, et convallis massa.

Vestibulum et finibus massa. Donec id fermentum velit. Curabitur at ipsum sit amet risus suscipit scelerisque. In ultricies, odio at volutpat.

Visual Design – Orientation cues

- People with disabilities use visual and non-visual cues to orient themselves on web pages and applications
- Focus indicators show interactive elements
- Spacing and grouping communicate regions

Visual Design – Orientation cues – Examples



Click Me



Click Me

Visual Design – Orientation cues – Examples

Welcome to Our Website

Discover Amazing Content Here

[Home](#) [About](#) [Services](#) [Contact](#)

Main Content

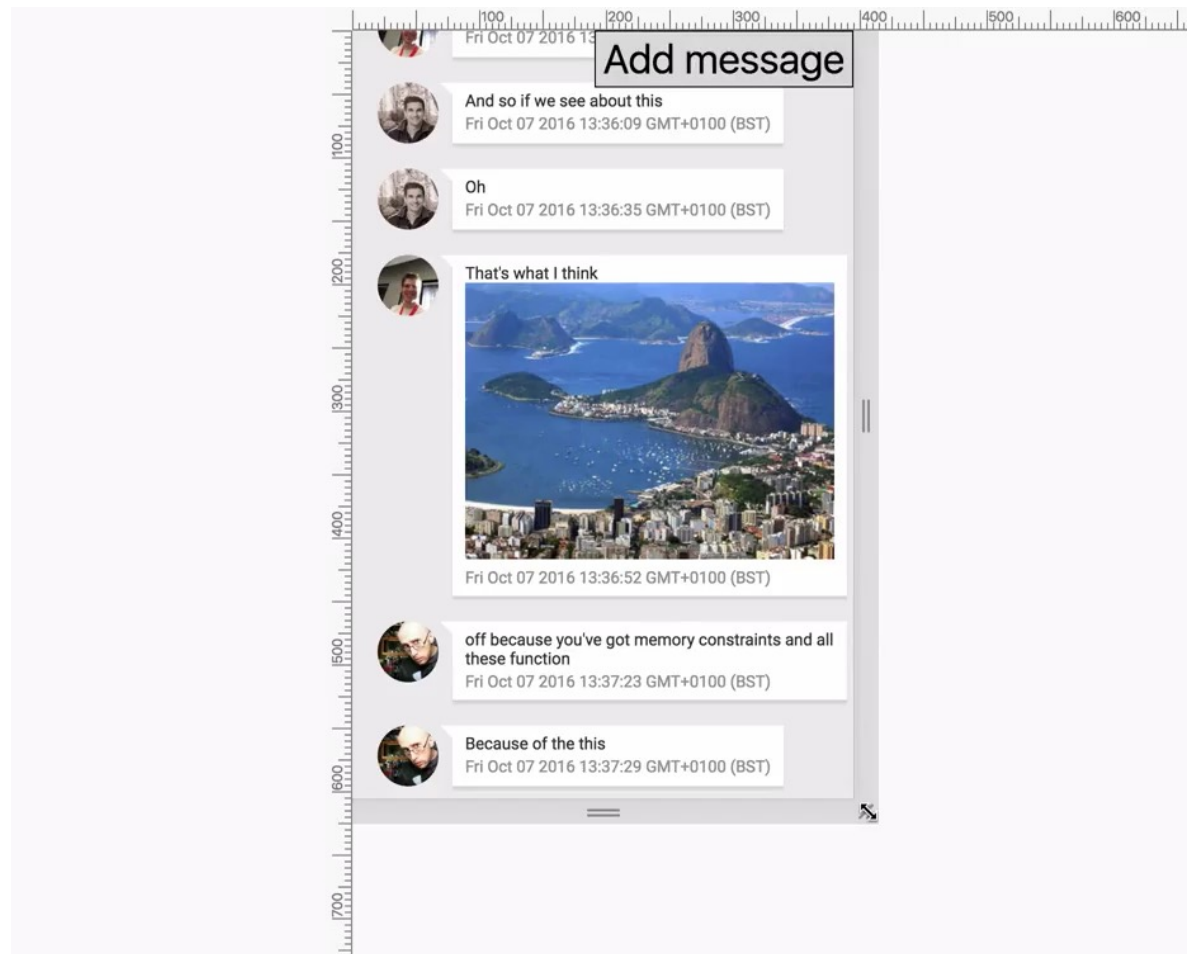
This is the main content area where you'll find all the exciting information.

© 2023 Our Website. All rights reserved.

Visual Design – Flexible layouts

- Responsive design
- Consider how content sequence, focus indicators, and target size can be designed to adapt to different screen sizes and screen configurations

Visual Design – Flexible layouts – Example



Information Design – Text and structure



- Use text and semantic structure to communicate the organization of pages and documents
 - Pages to split the content into logical units
 - Landmarks to define page regions
 - Headings with their corresponding rank levels to group related information.
- Use diagrams and illustrations to help users understand the information provided in text
- Provide specific definitions of words and meaning of phrases, such as professional terms, idioms, and jargon
- Provide the expanded form or meaning of acronyms and abbreviations

Information Design – Text and structure – Examples

ClickHelp

Product ▾ Use Cases ▾ Our Customers ▾ Blog Pricing 🔍 [Free Trial](#) [Request Demo](#)

Tools for Keywords

Keyword research tools can give the following information: if users search for this particular keyword, the number of its average monthly searches, the number of websites that get ranked under this keyword, etc. There are lots of different tools for keywords, the information they give can be a bit different, so it makes sense to combine several tools at a time. Let's have a look at the list of the most popular ones:

Follow Us:



Google Keyword Planner

Google Keyword Planner is a free tool, but you can't use it until you have a Google Ads account. With the help of this tool, you can discover new keywords and keyword ideas, get the search volume and forecasts for the keywords, compare trends, and study the historical stats for keywords.

Ahrefs Keywords Explorer

Ahrefs is one of the leading SEO tools with a wide variety of features. It may help you not only research keywords but also analyze backlinks and audit your website for SEO and many more things. You don't have to be an SEO expert to use this tool. Ahrefs is paid, even the trial period is not free.

Semstat

Information Design – Naming and grouping

- Use labels, instructions, and graphical representations to help people understand the purpose and functionality of user interface components.

Information Design – Naming and grouping - Examples

Form **Mail** Messages Additional Settings

Form

You can edit the form template here. For details, see [Editing Form Template](#).

text email URL tel number date text area drop-down menu checkboxes radio buttons Honeypot acceptance quiz file submit

```
<label> Your Name (required)
  [text* your-name] </label>

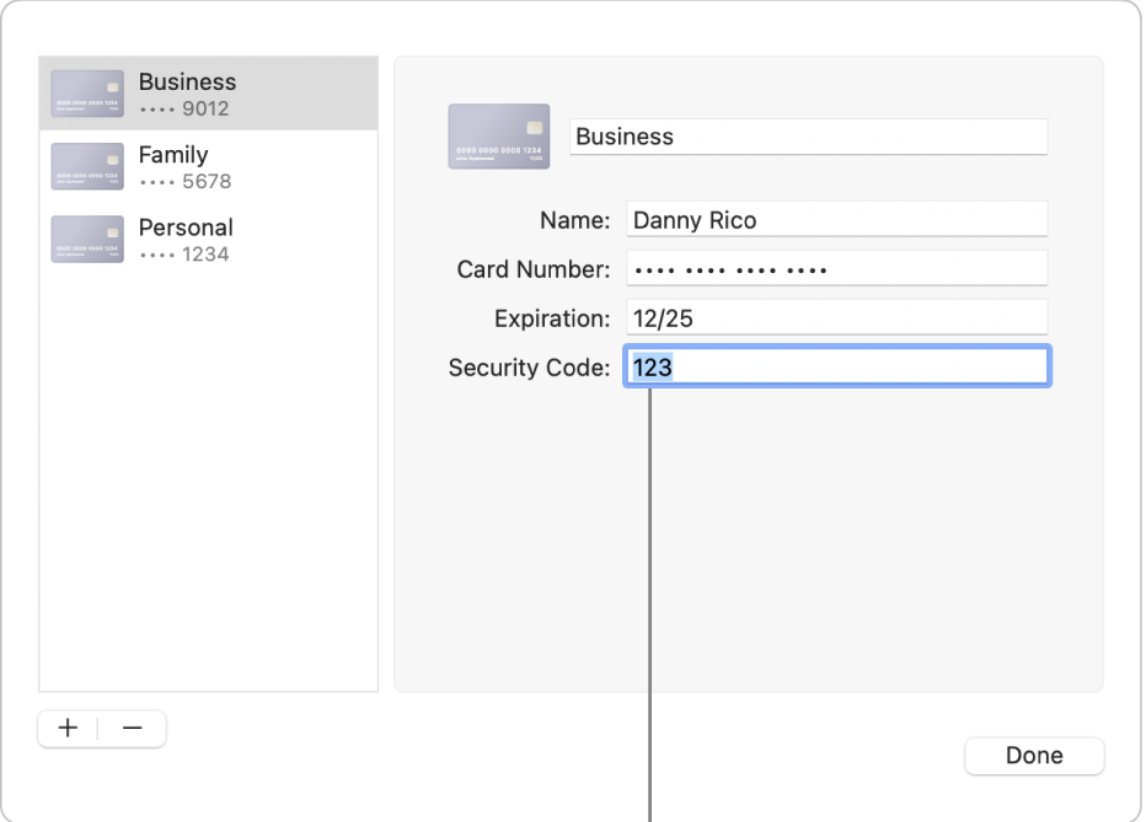
<label> Your Email (required)
  [email* your-email] </label>

<label> Subject
  [text your-subject] </label>

<label> Your Message
  [textarea your-message] </label>

[submit "Send"]
```

Information Design – Naming and grouping - Examples



Business
**** 9012

Family
**** 5678

Personal
**** 1234

Business

Name: Danny Rico

Card Number: **** * * * * *

Expiration: 12/25

Security Code: 123

+

-

Done

Save the security code with a credit card (optional).

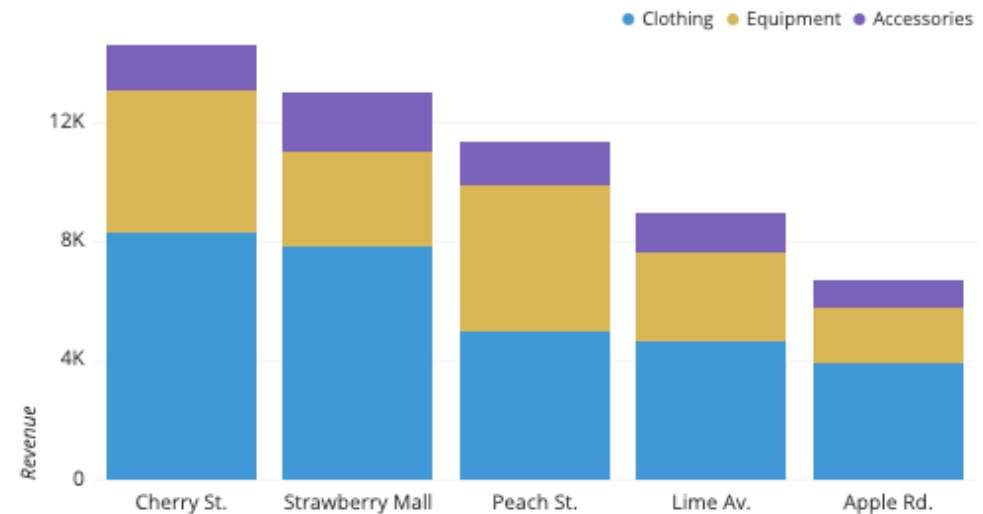
Information Design – Tables

- Use textual descriptions, graphical representations of the table content, as well as several simpler tables or individual lists to present the information in a complex table

Information Design – Tables – Examples

Product	Category	Price	Quantity
Widget A	Electronics	10.00	50
Widget B	Electronics	15.00	25
Widget C	Clothing	20.00	10
Widget D	Clothing	30.00	5

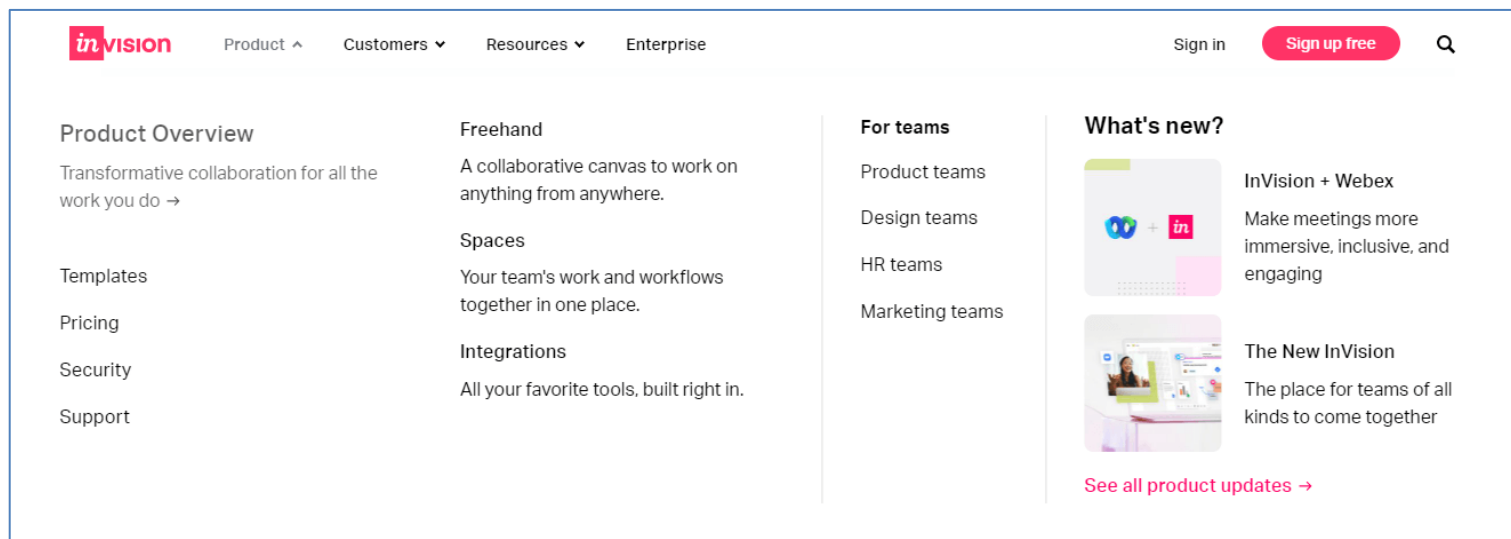
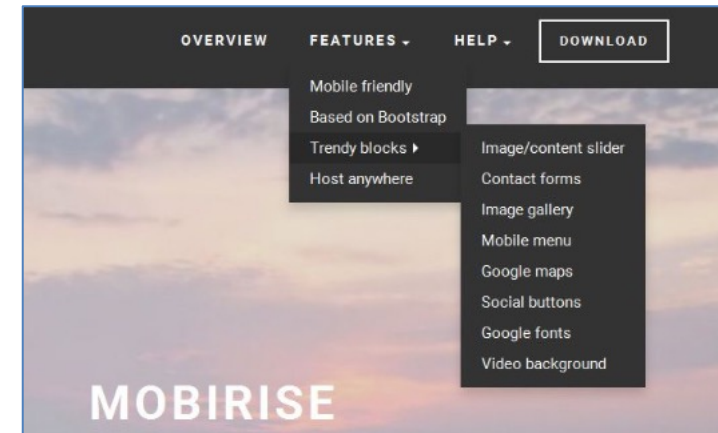
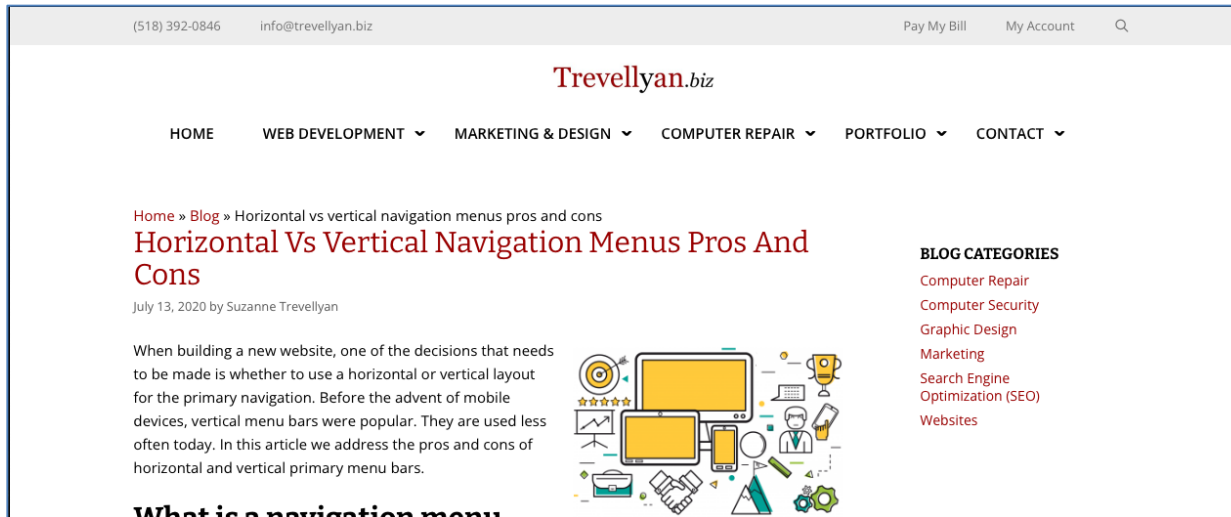
 Export to Sheets



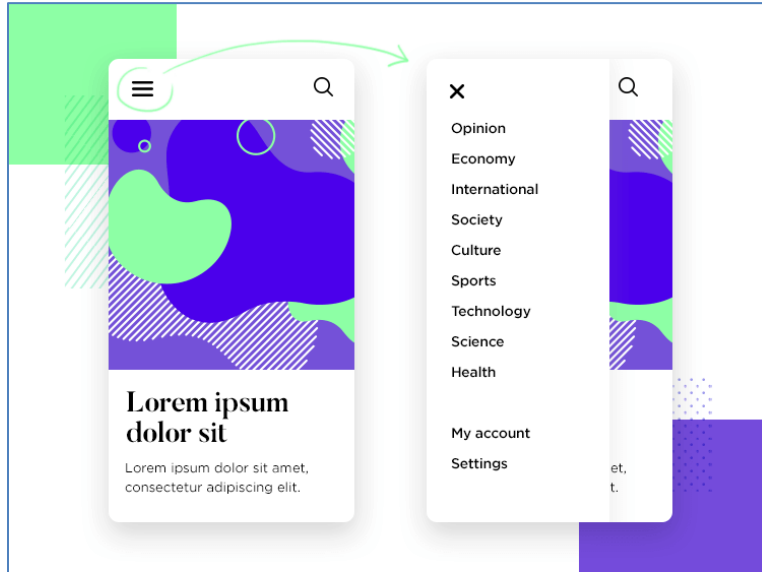
Navigation Design – Menus

- Consider different types of navigation menus, for example static, fly-out, and mega menus.
- Consider styling conventions for menus, such as expected location, appropriate size, and ability for menus and menu items to resize depending on different user configurations.

Navigation Design – Menus – Examples



Navigation Design – Menus – Examples



ADMINISTRATOR FEATURES

- Beverage List <
- Custom Lists **New** <
- Print Your Menu
- Website Menu
- Events
- Social Media **New** <
- Digital Menu
- Inventory <
- Analytics <
- POS Settings **New**

CUSTOMIZE YOUR WEB WIDGET

[Save Menu Configuration & Reload](#)

Menu Options

Name* Full Beverage Menu
This is not displayed anywhere on your menu

Display Beverages

- ☒ Beer on Tap & Cask
- ☒ Beers in Bottles & Cans
- ☐ Beers On Deck
- ☒ Wine
- ☒ Cocktails
- ☒ Spirits
- ☐ Non-Alcoholic
- ☐ Events

Display Menu Items

- ☐ Coffee Menu
- ☐ Concentrate
- ☐ Evergreen Flower Menu
- ☐ Flower
- ☐ Lunch Menu
- ☐ Merchandise
- ☐ Sandwich Menu
- ☐ Snack Menu
- ☐ Snacks

Beer On Tap & Cask Display Options

Order by* Name

Display Fields

- ☐ Exclude Brewery info
- ☒ Prefer brand logos
- ☐ Food Pairings
- ☐ Highlight Independent Breweries

Description No Description

Show RateBeer Scores Do not show scores

Beer In Bottles & Cans Display Options

Order by* Name

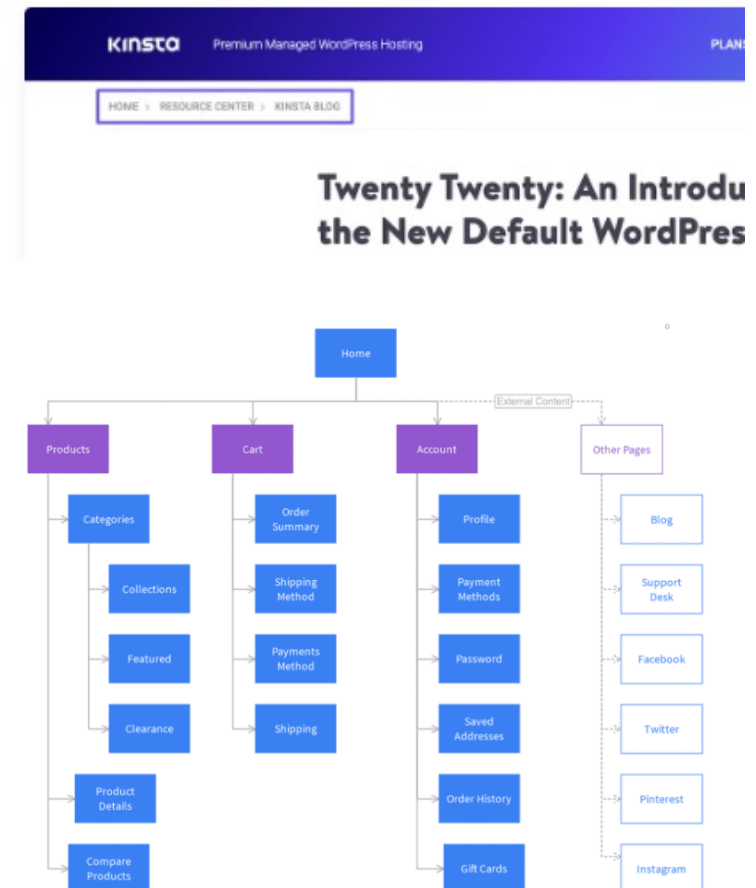
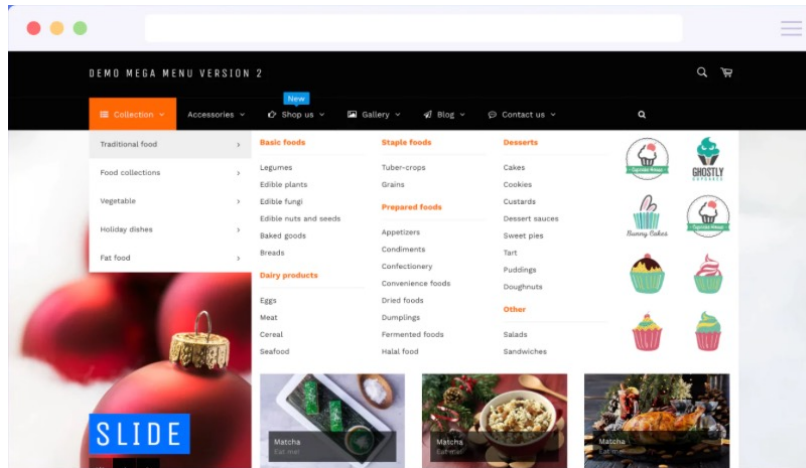
Display Fields

- ☐ Exclude Brewery info
- ☒ Prefer brand logos
- ☐ Food Pairings
- ☐ Highlight Independent Breweries

Navigation Design – Site Navigation

- Strategies people use to navigate to different pages of the same site:
 - Menus
 - Breadcrumb trails
 - Search
 - Site maps
- Different people rely on different navigational components

Navigation Design – Site Navigation – Examples

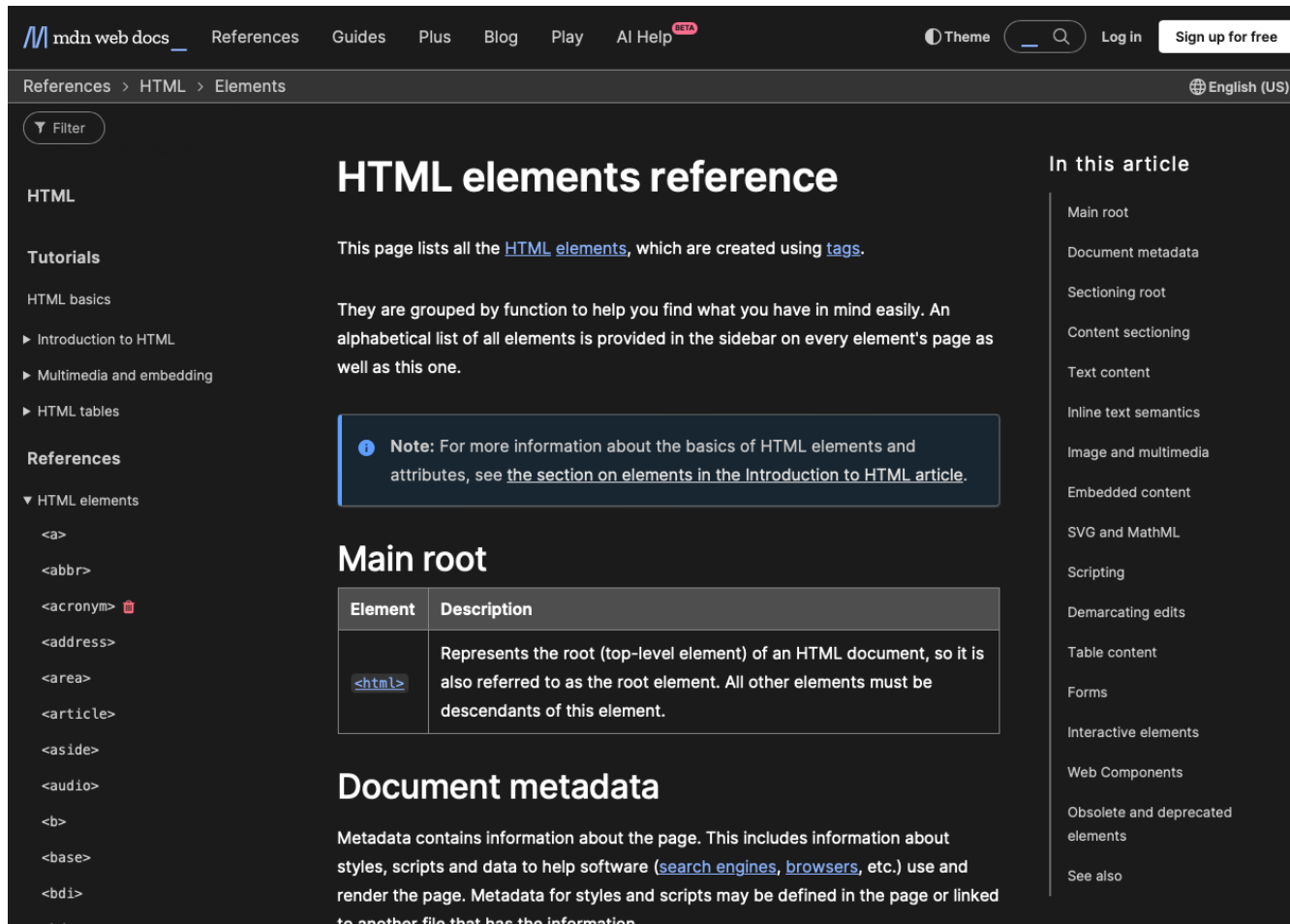


Navigation Design – Page Navigation



- Strategies that people use to navigate to different parts of a web page:
 - Tables of contents
 - Methods to bypass blocks of repeated content
- Different groups of users rely on different navigational components

Navigation Design – Page Navigation - Examples



The screenshot shows the MDN Web Docs 'HTML elements reference' page. The page has a dark theme and a top navigation bar with links to 'References', 'Guides', 'Plus', 'Blog', 'Play', and 'AI Help'. A search bar and 'Log in' button are also present. The breadcrumb trail is 'References > HTML > Elements'. The left sidebar contains a 'Filter' button and a list of categories: 'HTML', 'Tutorials', 'References', and 'HTML elements'. The main content area is titled 'HTML elements reference' and includes an introductory paragraph, a note about the basics of HTML elements, and a section for 'Main root' with a table. The right sidebar, titled 'In this article', lists various HTML features like 'Main root', 'Document metadata', 'Sectioning root', etc.

mdn web docs References Guides Plus Blog Play AI Help BETA Theme Search Log in Sign up for free

References > HTML > Elements English (US)

Filter

HTML elements reference

This page lists all the [HTML elements](#), which are created using [tags](#).

They are grouped by function to help you find what you have in mind easily. An alphabetical list of all elements is provided in the sidebar on every element's page as well as this one.

Note: For more information about the basics of HTML elements and attributes, see [the section on elements in the Introduction to HTML article](#).

Main root

Element	Description
<html>	Represents the root (top-level element) of an HTML document, so it is also referred to as the root element. All other elements must be descendants of this element.

Document metadata

Metadata contains information about the page. This includes information about styles, scripts and data to help software ([search engines](#), [browsers](#), etc.) use and render the page. Metadata for styles and scripts may be defined in the page or linked to another file that has the information

In this article

- Main root
- Document metadata
- Sectioning root
- Content sectioning
- Text content
- Inline text semantics
- Image and multimedia
- Embedded content
- SVG and MathML
- Scripting
- Demarcating edits
- Table content
- Forms
- Interactive elements
- Web Components
- Obsolete and deprecated elements
- See also

Navigation Design – Page Navigation - Examples

[Skip to main content](#)



 [Menu](#)

[Home](#) / [Solution home](#) / [Guides & FAQs](#) / [Accessibility](#) / **How skip links can help with website accessibility**

How skip links can help with website accessibility

Modified on: Tue, 28 Sep, 2021 at 5:09 PM

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Interaction Design – Keyboard Interactions

- Support standard keyboard interactions, for example the use of the tab, enter, escape, and arrow keys
- Keyboard support benefits other ways of interacting, including speech interaction and effective use of some assistive technologies
- Custom keyboard interactions can support efficiency but can also disrupt the user's expectations. Designers must document these custom interactions and make them consistent throughout the interface.

Interaction Design – Keyboard Interactions



Ciências
ULisboa

Keyboard shortcuts Open in a new window | Close

Compose and Chat

- Shift + Esc**: Focus main window
- Esc**: Focus latest chat or compose
- Ctrl + ,**: Advance to next chat or compose
- Ctrl + ,**: Advance to previous chat or compose
- % + Enter**: Send
- % + Shift + c**: Add Cc recipients
- % + Shift + b**: Add Bcc recipients
- % + Shift + f**: Access custom from
- % + Shift + d**: Discard draft
- % + k**: Insert a link
- % + ;**: Go to previous misspelt word
- % + '**: Go to next misspelt word
- % + Shift + !**: Open spelling suggestions

Navigating within the sidebar in Google Chat

- % + o**: Browse spaces
- % + i**: Create a space
- ↑**: Focus on previous space or direct message
- ↓**: Focus on next space or direct message
- Shift + ↑**: Focus on previous unread space or direct message
- Shift + ↓**: Focus on next unread space or direct message
- % + Shift + k**: Switch spaces or direct messages quickly
- or Enter**: Select space or direct message

Navigating messages within a thread in Google Chat

- ↑**: Focus on previous message
- ↓**: Focus on next message
- Shift + ↑**: Focus first message in thread
- Shift + ↓**: Focus last message in thread
- ↑**: Go back to main conversation
- Esc**: Exit to thread navigation panel
- r**: Focus on reply box

Reply box in Google Chat

- ↑**: Edit previously posted message
- Shift + Enter**: New line
- ⌘ + Enter**: Show emoji picker

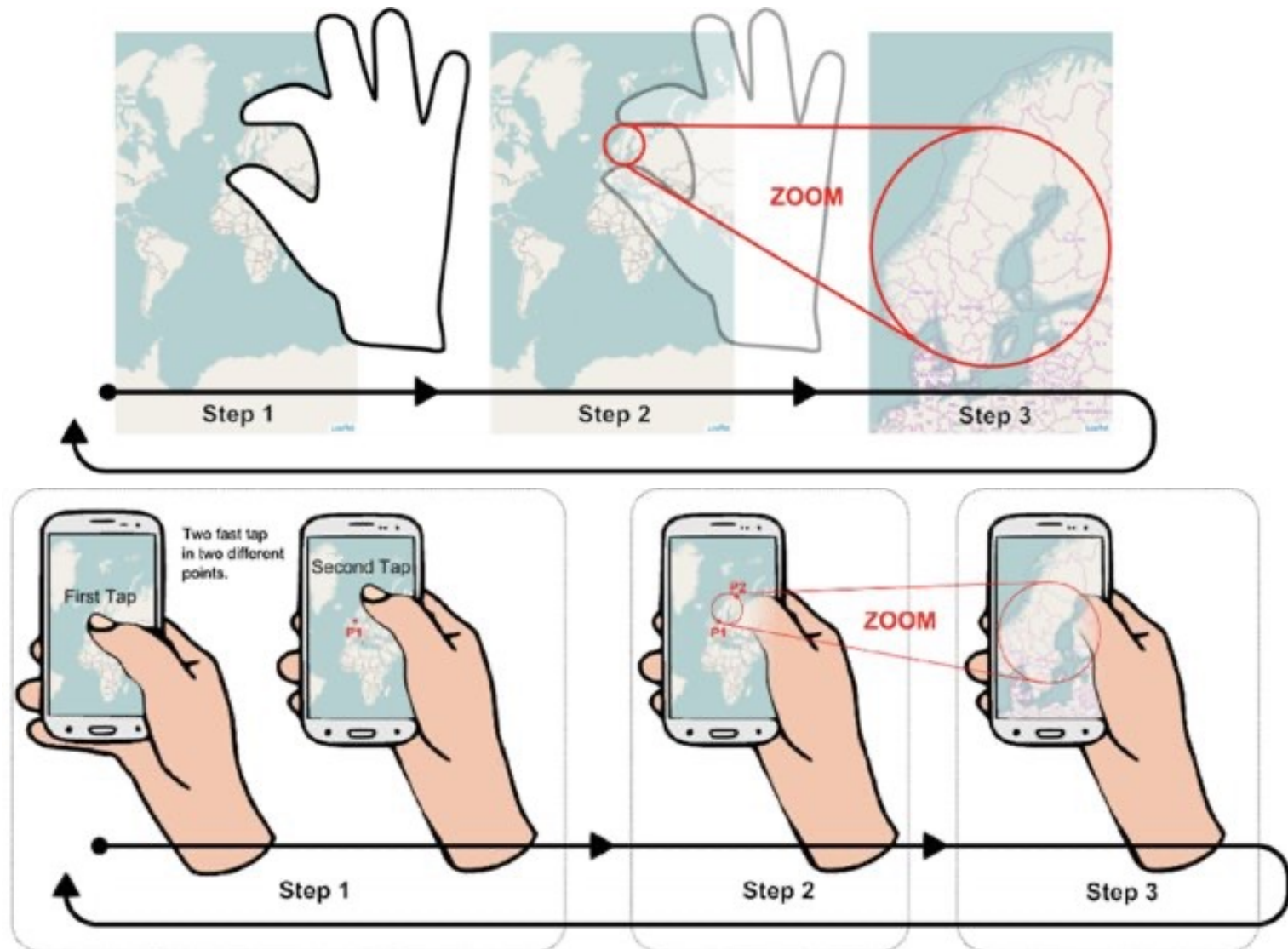
Formatting

- % + z**: Undo
- % + y**: Redo
- % + Shift + 5**: Previous font
- % + Shift + 6**: Next font
- % + Shift + -**: Decrease text size
- % + Shift + +**: Increase text size
- % + b**: Bold
- % + i**: Italics
- % + u**: Underline

Interaction Design – Gestures and Motion

- User interfaces can react to motion, such as shaking the device to perform a specific action. Some people may perform these motion-based gestures inadvertently, while others may not be able to perform them at all.
- Present alternatives to motion-based gestures, for example providing methods to undo, abort, and reverse actions performed using complex gestures.

Interaction Design – Gestures and Motion Examples



Images and Graphics – Functional Images

- Functional images convey the purpose of a component
 - Search, print, and save functionality are often presented using icon imagery
- These images need an equivalent, succinct text alternative that explains the function rather than a description of the image

Images and Graphics – Functional Images – Examples



Images and Graphics – Informative Images

- Informative images convey information that is necessary to understand the content.
- Their description generally requires a short phrase or sentence.

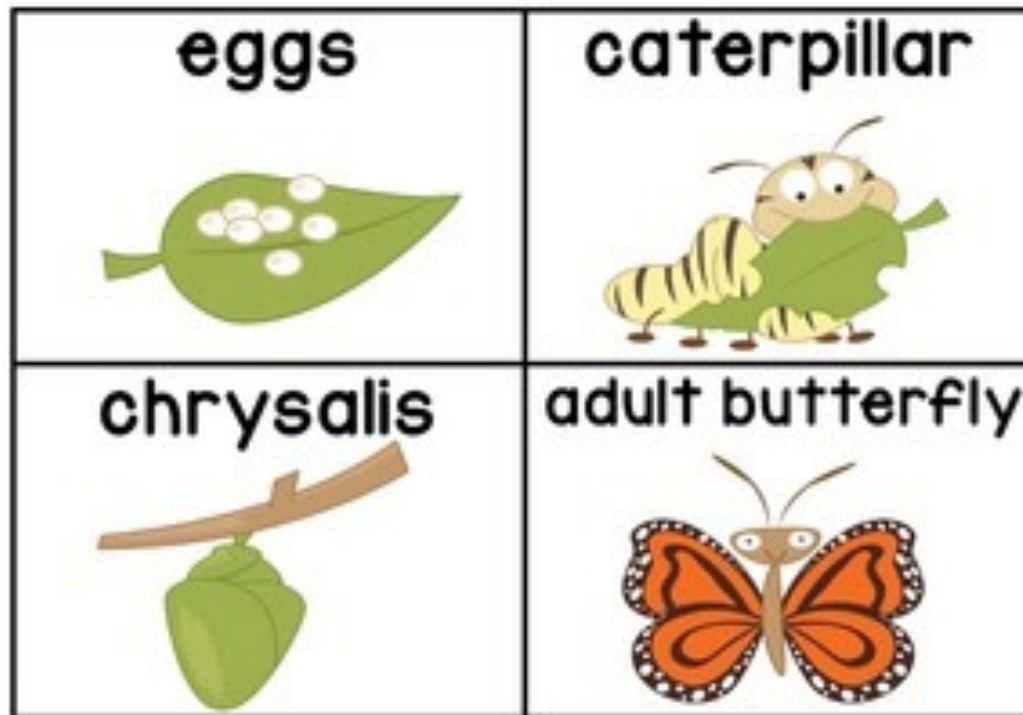
Images and Graphics – Informative Images

Examples



Images and Graphics – Informative Images Examples

Butterfly Life Cycle

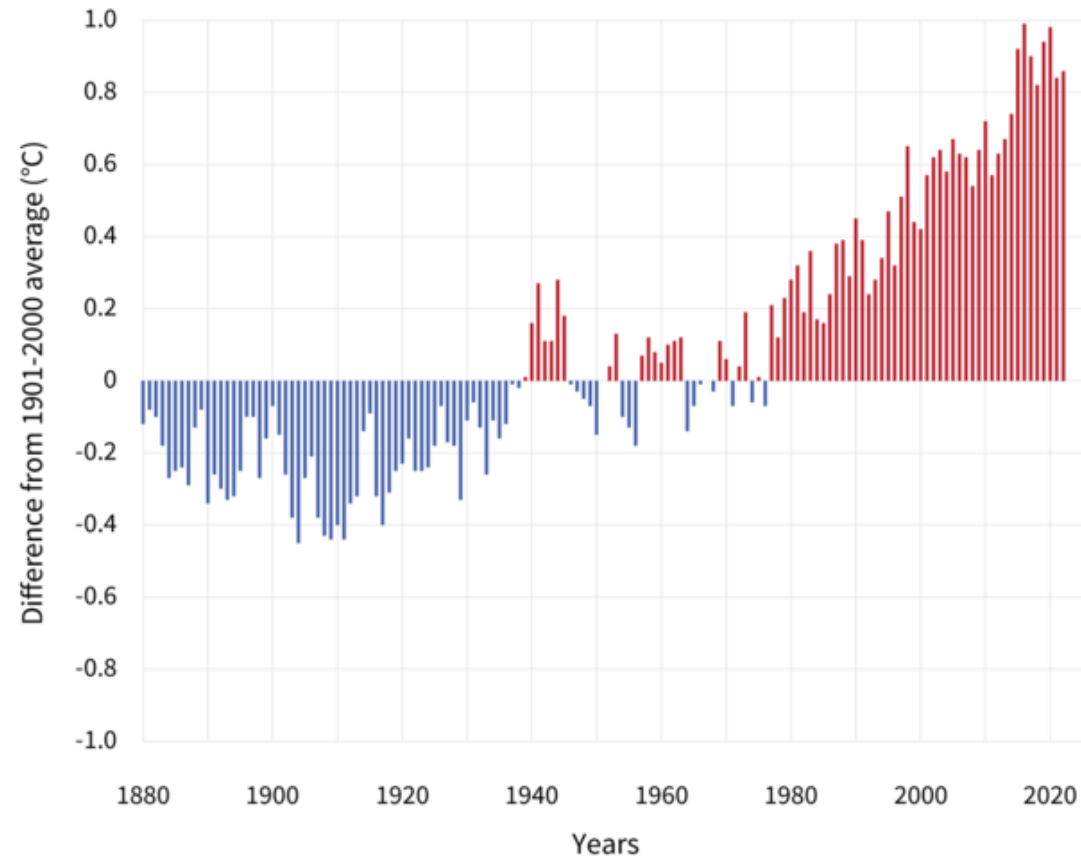


Images and Graphics – Complex Images

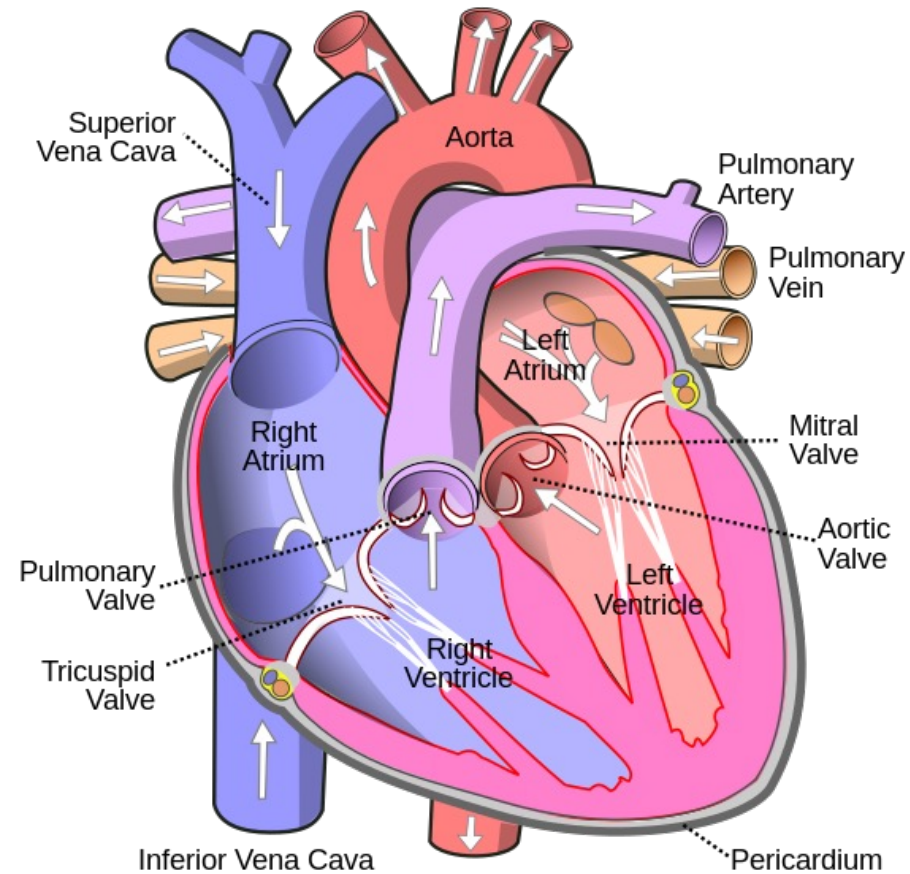
- Complex images are those whose description requires more than a short phrase or sentence

Images and Graphics – Complex Images – Examples

GLOBAL AVERAGE SURFACE TEMPERATURE



Images and Graphics – Complex Images – Examples



Images and Graphics – Decorative Images

- Decorative images are used for ornamental purposes and convey no information or function
- Decorative images require empty text alternatives, so that they can be ignored by assistive technologies, and reduce noise

Images and Graphics – Decorative Images -



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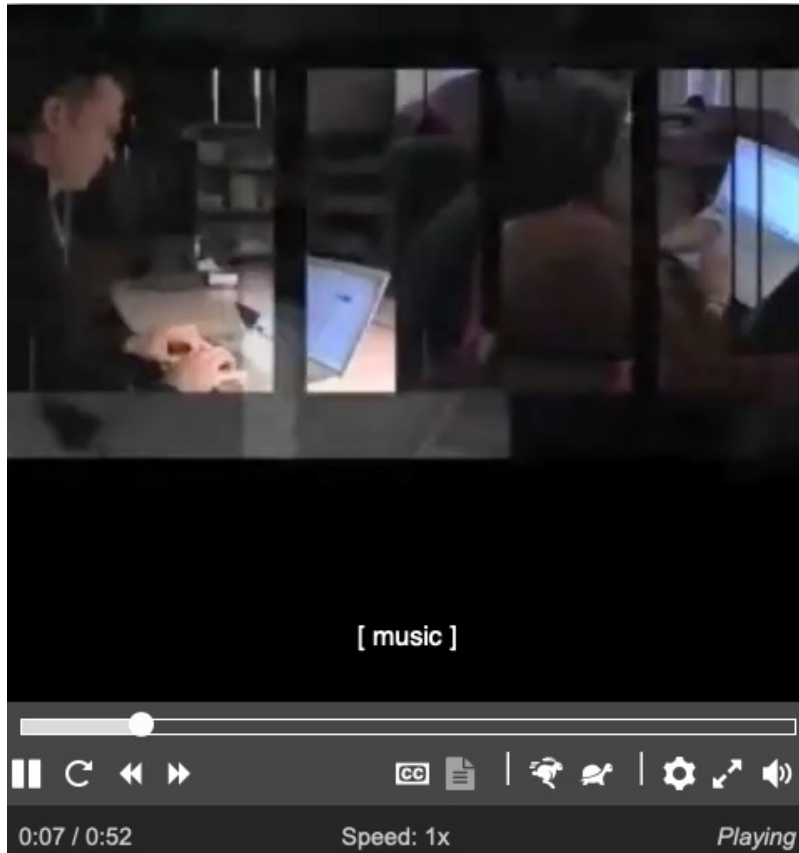


Multimedia and Animations – Alternatives to multimedia

- Descriptions are essential for people who don't see visual media content
- Captions are essential for people who don't hear audio content
- Text transcripts provide an alternative to both visual and audio information
- Plan in the design phase to accommodate alternatives to multimedia content such as description, captions, text transcripts, and sign languages
- It is generally more reliable to use an existing media player with support for accessibility than to design one from scratch
 - Accessible media players are operable by keyboard, make keyboard focus visible, have clear labels, have appropriate contrast ratios, and are easy to find on the page

Multimedia and Animations – Alternatives

multimedia – Examples



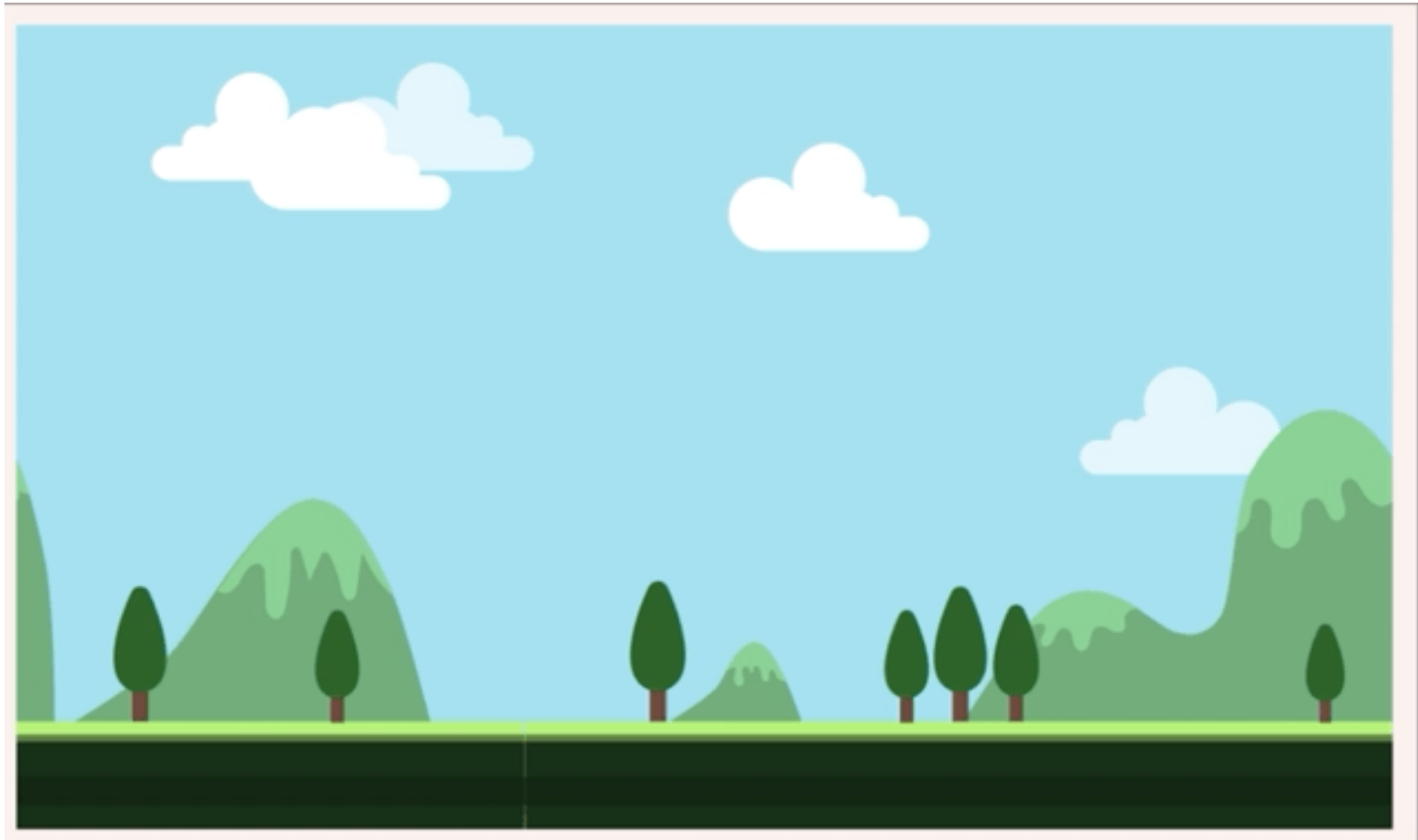
Able Player

<https://ableplayer.github.io/ableplayer/>

Multimedia and Animations – Movement and Animations

- Types of moving content: carousels, animations and others
- Avoid seizures, physical reactions, distractions, and unpredictable keyboard focus management caused by moving content
 - For example, support for pausing, stopping, or hiding moving content, as well as minimizing blinking and flashing

Multimedia and Animations – Movement and Animations – Examples



INFORMATION ARCHITECTURE

Information Architecture

- For as long as people have had information to convey, they have had to make choices about how they structure that information so other people can understand and use it.
- Information architecture is closely related to the field of information retrieval – the design of systems that enable users to find information easily.

Definition

1. The **structural design** of shared information environments
2. The combination of **organization, labeling, search, and navigation** systems within web sites and intranets
3. The art and science of **shaping** information **products** and **experiences** to support **usability and findability**

Inherent challenges

- No document fully and accurately represents the intended meaning of its author.
- No label or definition totally captures the meaning of a document.
- No two readers experience or understand a particular document or definition or label in quite the same way.

IA concepts

Structuring – determining the appropriate levels of granularity for the information “atoms” in your site and deciding how to relate them to one another.

Organizing – involves grouping those components into meaningful and distinctive categories.

Labeling – figuring out what to call those categories and the series of navigation links that lead to them.

Why is it important?

- The cost of finding information
- The cost of not finding information
- The cost of construction
- The cost of maintenance
- The cost of training

- Why do users come to a site? They have an information need!
- Information needs can vary widely, and each type of information need causes users to exhibit specific information-seeking behaviors.

Filtros da Pesquisa

Nome: Departamento:

Extensão: Sala:

Procurar

Pub



[Limpe o seu Mac](#)

Folhas de cálculo

Demonstrações Financeiras SNC
Análise Financeira
Análise de Projectos de Investimento
Gua de Implementação do Balanced Scorecard
Previsão de Tesouraria
Cálculo de Rendas Financeiras
Todas as folhas de cálculo

Os mais lidos hoje



Groupon: sim ou não?
Escrito por Steve Strauss
O meu colega gostava que nos juntássemos ao popular Groupon com a nossa loja de bicicletas para fazer uma promoção...



Oito Perguntas que Todos os Empresários Deviam Perguntar a Si Próprios
Escrito por Joshua Kennon

Se eu estivesse a pensar em começar uma pequena empresa do nada ou se já estivesse a operar uma pequena...



As 12 Consequências de Perceber que Não Há Bolsa de Valores

Bolsa de Valores Portugal

Acções, Divisas, Petróleo, Ouro Software Gratis e Bónus €25 Gratis!

Anúncios Google

419

Recommend

Fórmula de Valor Intrínseco de Benjamin Graham



Uma análise sobre como o pai do investimento de valores calculou o valor intrínseco de...

Continuar...

O índice de acções inevitáveis



Fiquei curioso sobre o que seria necessário para construir um índice de acções ordinárias que...

Continuar...

Qual a verdadeira rentabilidade dos certificados do tesouro?

Com a crescente necessidade de obtenção de financiamento, a República Portuguesa criou os Certificados do...

Continuar...

Uma técnica para comparar o valor intrínseco de duas acções

Uma técnica que considero que ajuda

Acções vs. Obrigações vs. Valorização do Ouro nos Últimos 200 Anos

Há uma razão pela qual tendo a ser mais favorável aos investimentos em participações no...

Continuar...

As 12 Consequências de Perceber que Não Há Bolsa de Valores



Uma das coisas que me ajudou a construir riqueza, no início da minha carreira, foi...

Continuar...

Conhecimentos básicos para analisar um portefólio de investimento em acções

Pensei que fosse útil demonstrar como analiso um portfolio de investimento e calcular uma estimativa...

Continuar...

Procurar o valor de uma empresa através da análise do balanço

O balanço é a peça

- Searching for something you know is a very different behavior than browsing for the unknown.
- Distinguishing between these needs and behaviors and determining which are your users' highest priorities is an extremely valuable pursuit.

Fundamental questions

- What type of information is the user looking for?
- How much information is enough?
- How will users interact with your information architecture?

Information needs

- “The right answer”
 - What is the population of Portugal?
- “No answer”
 - I’d like to know about restaurants in Lisbon

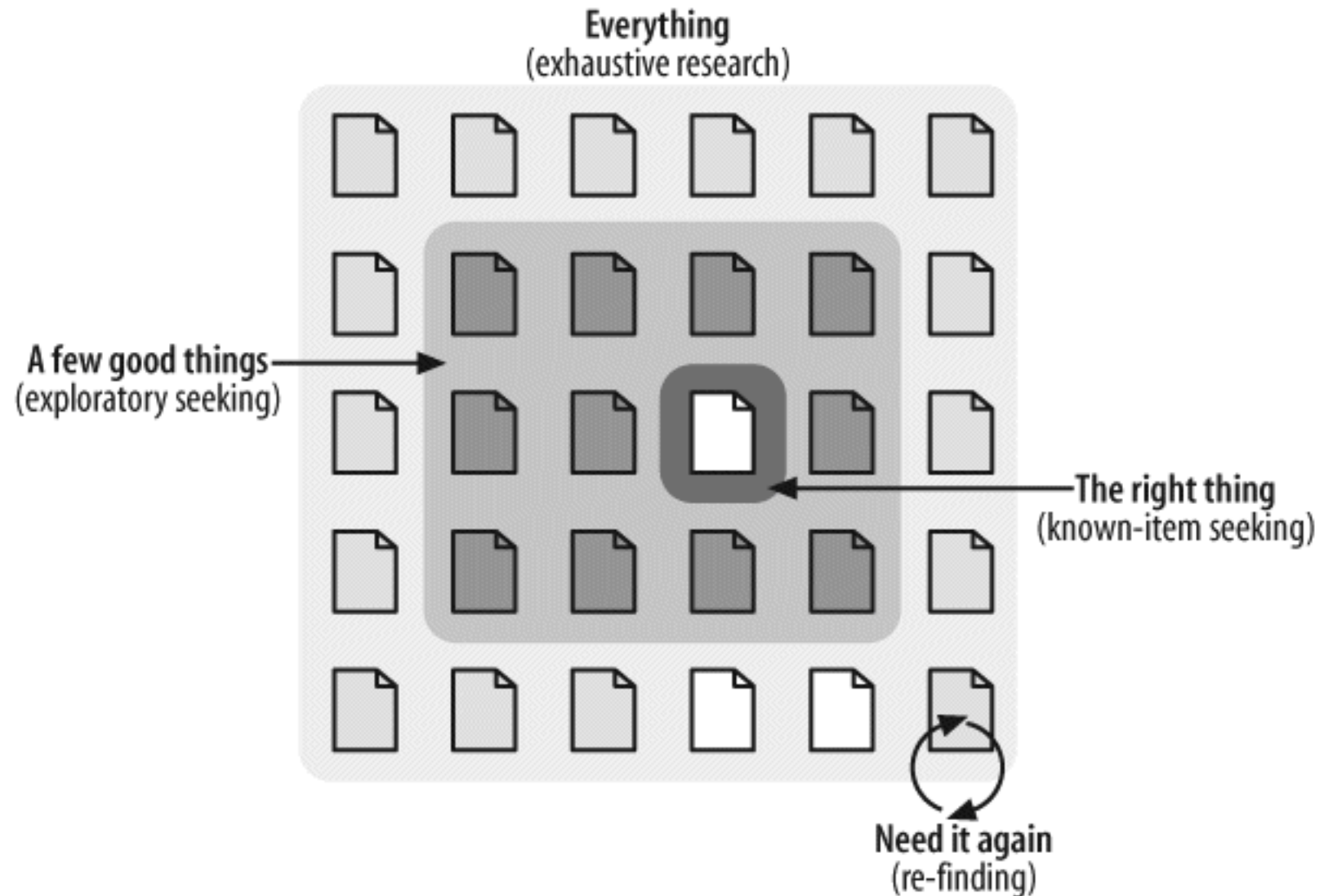
Information needs

- Known-item seeking
 - you usually know what you’re looking for, what to call it, and where you’ll find it
 - Ex.: finding out the phone number of a colleague
- Exploratory seeking
 - The user is not exactly sure what they are looking for
 - They are looking to learn something from the process of searching and browsing
 - There is no clear expectation of a “right” answer, nor does the user necessarily know how to articulate what exactly they are looking for
 - They are happy to retrieve a few good results and use them as a springboard for the next iteration of the search. It’s not always possible to definitively determine when exploratory searching is finished
 - Ex.: finding a good hotel in Oporto

Information needs

- Exhaustive research
 - The user is looking for everything on a particular topic
 - The user often has many ways to express what they are looking for and may have the patience to construct their search using all those varied terms.
 - Ex.: trying to learn more about a family member's medical condition
- Re-finding
 - Tag a piece of information to make it easy to find later on
 - Ex.: bookmark an interesting page for reading later

Information needs



Seeking behaviors

- Search
 - Browse
 - Ask
-
- We often **integrate** searching, browsing, and asking in the same finding session.

Seeking behaviors

- Our information architecture needs to be ready to adapt to changing user behaviors

Seeking behaviors



UNIVERSIDADE DE LISBOA
Faculdade de Ciências

Course categories



[First-cycle in Informatics Engineering](#)



[First-cycle in Information and Communications Technologies](#)Search courses:

Seeking behaviors

Google scholar information architecture Search Advanced Scholar Search

Scholar Articles and patents anytime include citations Create email alert

[BOOK] [Information architecture for the World Wide Web.](#)
L Rosenfeld... - 1998 - [eric.ed.gov](#)
O'Reilly & Associates, Inc., 101 Morris St., Sebastopol, CA 95472; toll-free phone: 800-998-9938; World Wide Web: <http://www.ora.com> (\$24.95). ... Help ERIC expand online access to documents currently available only on microfiche. Learn more about our efforts.
[Cited by 789](#) - [Related articles](#) - [Cached](#) - [All 11 versions](#)

[BOOK] [Information architecture for the world wide web](#)
L Rosenfeld... - 2002 - [books.google.com](#)
Praise for **Information Architecture** for the World Wide Web "It's been well worth the wait! This much expanded second version provides a holistic perspective on **information architecture**—something that wasn't possible earlier on when the concept was just ...
[Cited by 431](#) - [Related articles](#) - [All 12 versions](#)

[Information architecture: In search of efficient flexibility](#)
BR Allen... - MIS quarterly, 1991 - JSTOR
This article addresses how **information** systems **architecture** can be used to support organizations in the 1990s—organizations that face the dual challenge of "speed and flexibility" and "low cost and efficiency." At the heart of this challenge is the basic notion ...
[Cited by 27](#) - [Related articles](#) - [All 9 versions](#)

[HTML] [An architecture for information in digital libraries](#)
WY Arms, C Bianchi... - D-Lib Magazine, 1997 - [dlib.org](#)
Flexible organization of **information** is one of the key design challenges in any digital library. For the past year, we have been working with members of the National Digital Library Project (NDLP) at the Library of Congress to build an experimental system to organize and store ...
[Cited by 150](#) - [Related articles](#) - [Cached](#) - [All 6 versions](#)

[BOOK] [Information architecture for the world wide web](#)
P Morville... - 2006 - [books.google.com](#)
Preface The mind is slow to unlearn what it learnt early. —Seneca Since 1994, when we first began organizing web sites, we have enjoyed a rare opportunity to participate in the birth of a new discipline. In the early days, we were pioneers and evangelists, exhorting web ...
[Cited by 210](#) - [Related articles](#) - [All 27 versions](#)

Know the needs and behaviors

- Search analytics
 - Review the most common searches in the site
 - Know the needs (what is most searched)
 - Know the behaviors (how the searches are built)

THE ANATOMY OF AN INFORMATION ARCHITECTURE

Academics

Research

Student Life

Athletics

About Us

Prospective Students

Current Students

Faculty & Staff

Alumni

Parents & Family

Corporate Visitors

News & Events

[Brumley Receives Presidential Early Career Award](#)
Sept. 26

[Vote for Zach: Engineering the Best Mustard Around](#)
Sept. 26

[Dietrich's Unprecedented Gift to Advance How Students Create & Think Globally](#)
Sept. 14

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Sept. 17 - Dec. 11

Carrier 3:00 PM

Back Full Body Blaster 45

WORKOUT TIME: 45:00

Start Workout

WARM UP

Tap to show warmup
10 EXERCISES

STRENGTH TRAINING

Machine Shoulder Press
2 SETS / 10 REPS

Dumbbell Lunge (long stride)
2 SETS / 10 REPS

Dumbbell Bench Press
2 SETS / 10 REPS

Machine Pulldown
2 SETS / 10 REPS

Carrier 2:05 PM

Cancel New Workout

MINUTES

35 40 45

LOCATION Home

Gym

GOAL

Health & Longevity

Fat Loss

Muscle

FOCUS

Full Body

Inner Body

HEALTH & WELLNESS

Everything to GAIN

Working out but getting nowhere? No time or money for a personal trainer? Nicholas Gammell has your answer.

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Battery Bulk

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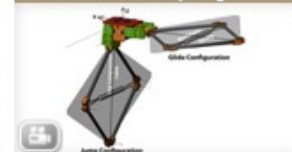


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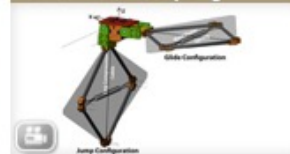


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Anticipate the needs

1. Where am I?
2. I know what I need. How do I search for it?
3. How do I get around this site?
4. What's important and unique about this organization?
5. What's available on this site?
6. What's happening here?
7. How can I contact a human?
8. What's their address?

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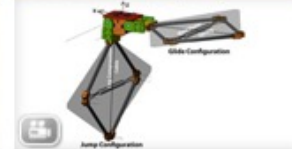


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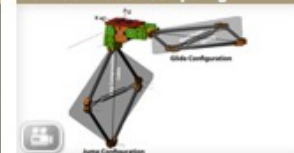


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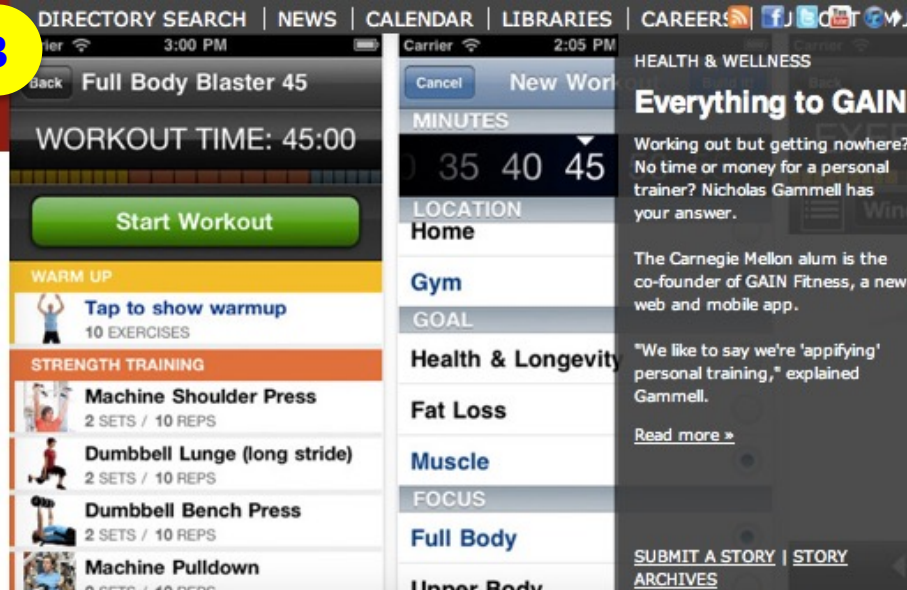
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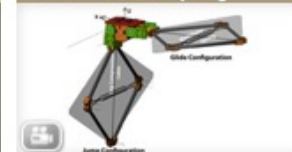


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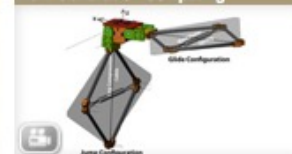


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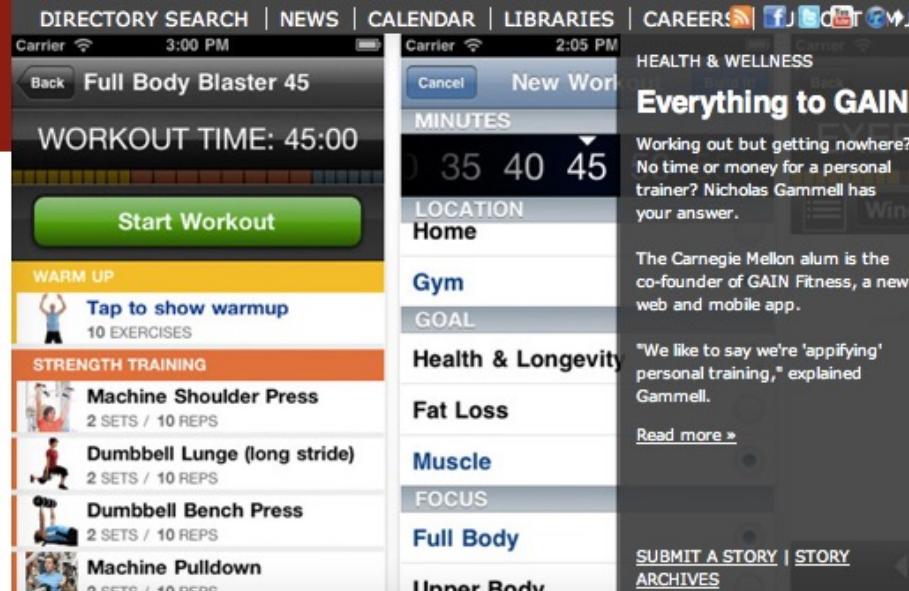
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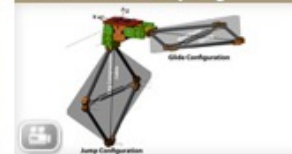


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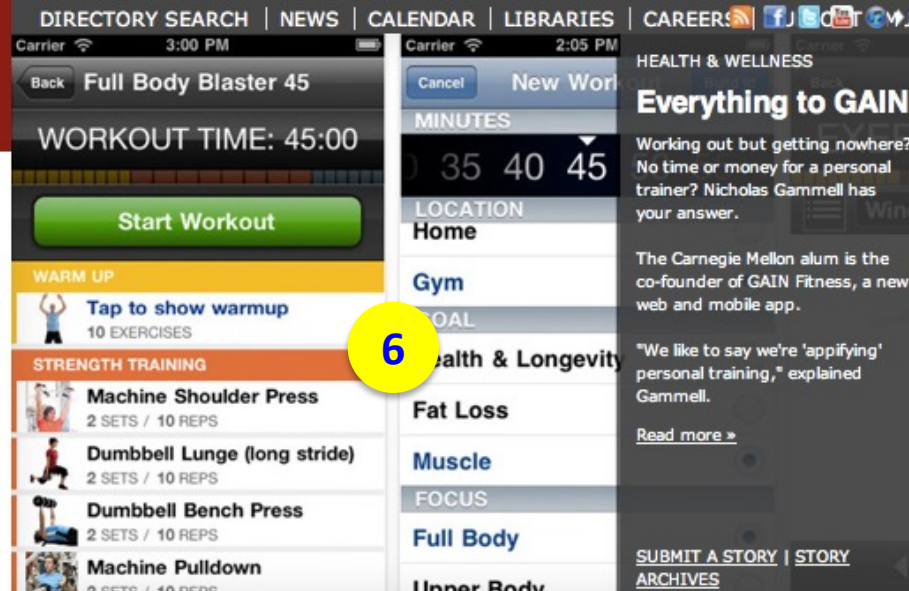
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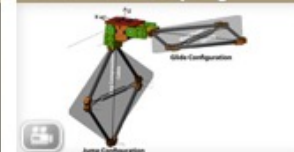


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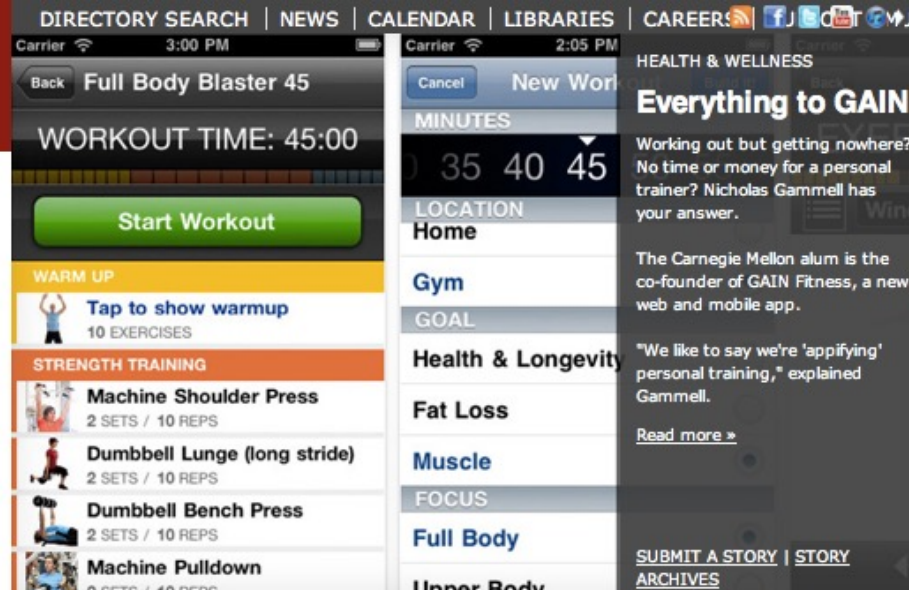
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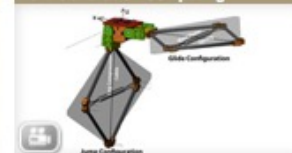


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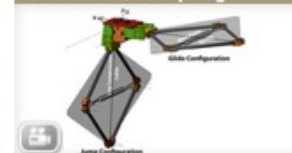


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Categorization schemes

- Most commonly, information architecture problems require creating categorization schemes that will correspond to our own objectives for the site, the user needs we intend to meet, and the content that will be incorporated in the site.
- But how are the categorization schemes created?

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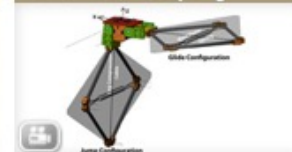


Legacy Lives On

Family, friends and the campus community united last weekend to remember beloved professor Lester Lave, a visionary researcher and treasured icon of Carnegie Mellon University. One of the nation's leading environmental economists, Lave and his legacy will be honored by the establishment of a new professorship and fellowship.

[Read more »](#)

Next Generation Computing



A Leap Ahead

The tiny robot jumps up, spreads its wings and glides away. Matthew Woodward's vision is moving closer to reality. The Carnegie Mellon University mechanical engineer has developed a miniature prototype — the first robot to incorporate two high-performance functions.

[Read more »](#)

A TRULY GLOBAL UNIVERSITY

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innovation.
THE CAMPAIGN FOR
CARNEGIE MELLON UNIVERSITY

Today
CARNEGIE MELLON

Visualizing information architecture



- This is *top-down information architecture*

Cozido à Portuguesa

Doses: 4

Modo preparação: Lume

Tempo preparação: 120

min

Grau dificuldade: Médio



Carne de vaca, adequada para cozer
meia galinha;
1 pé de porco, entrecosto, chispe;
presunto, chouriço, farinheira, salpicão;
toucinho salgado, bacon;
orelheira fresca e fumada;
couve portuguesa (penca) ou coração;
cenouras, batatas, nabos;
sal e azeite q.b.



Média de quantidades
por pessoa



Tabela de conversão das
medidas utilizadas



1. Comece por cozer todas as carnes numa panela grande. As carnes que forem mais salgadas devem ficar de molho previamente umas horas e só depois se colocam a cozer.
2. Regue a água da cozedura com um fio de azeite e tempere a gosto. Por ordem de cozedura mais rápida, vão-se tirando os enchidos, depois as carnes de porco, e só no fim é que se retira a carne de vaca depois de bem cozida.
3. Aproveitando a mesma água de cozer as carnes, coloque os legumes. Quando estes estiverem cozidos, retire a panela do lume, deixando os legumes dentro.
4. Para servir, corte as carnes, disponha numa travessa com os respectivos legumes. Pode acompanhar com feijão branco cozido, também cozido na água dos legumes, e arroz branco.

Visualizing information architecture

- Where is the information architecture?
 - Navigation options
 - Title
 - Ingredient list
 - Preparation instructions
 - Supports searching (I can search for the recipe's name)
 - Supports browsing (it's organized like a recipe)

Visualizing information architecture



- This is *bottom-up information architecture*
- Important because users are increasingly likely to bypass your site's top-down information architecture (they don't arrive at the home page when using web-wide search tools)

Categorization schemes

- Neither approach is better than the other.
 - Approaching the architecture from the top down can sometimes cause important details about the content itself to be overlooked.
 - A bottom-up approach can sometimes result in an architecture so precisely tuned and fitted to the existing content that it isn't flexible enough to accommodate changes or additions.

Information architecture components

- Organization systems
- Labeling systems
- Navigation systems
- Searching systems

ORGANIZATION SYSTEMS

Organization

- Socially we organize information to understand, explain and control
 - Where do we live
 - Street < City < District < State < Country < Continent
 - Where do we work
 - Department < Company < Industry
- As information architects, we organize information so that people can find the right answers to their questions

Challenges

- Ambiguity
 - e.g. pitch
 1. A throw, fling, or toss
 2. A black, sticky substance used for waterproofing
 3. The rising and falling of the bow or stern of a ship in rough sea
 4. A salesman's persuasive line of talk
 5. An element of sound determined by the frequency of vibration
 6. ...

Challenges

- Ambiguity
 - When we use words as labels for our categories, we run the risk that users will miss our meaning.
 - We have to decide in which groups we place the words
 - Tomato - a red or yellowish fruit with a juicy pulp, used as a vegetable: botanically it is a berry (yourdictionary.com)

Challenges

- Heterogeneity
 - Websites often provide access to documents and their components at varying levels of granularity
 - Articles and journals and journal databases side by side
 - Links might lead to pages, sections of pages, or other websites
 - In the same website I can find financial news, product descriptions employee home pages, image archives and software files.
 - The characteristics of a document (journal) are not the same for its content (articles)
 - Each format will have uniquely important characteristics

Challenges

- Perspectives

- Examples:

- Each person organizes its file system in a unique and personal way
 - Each organization has a unique and different org chart for their departments and divisions

- We need to make an effort to understand how the user/client will search for the information

- To structure a website according to a company's org chart will not help a client looking for information on a product he wishes to buy!

Challenges

- Internal politics
 - In certain cases you must remind your colleagues or clients to focus on creating an architecture that works for the user. In others, you need to make compromises to avoid serious political conflict.

Organization

- Schemes
 - The shared characteristics of content items and the logical grouping of those items
- Structures
 - The types of relationships between content items and groups

Organization schemes

- Exact
 - Divide information into well-defined and mutually exclusive sections
 - Easy to design, maintain and use
 - E.g.: White Pages
 - Require users to know the specific name of the resource they are looking for
 - The white pages don't work very well if you're looking for a plumber!

Exact schemes

Alphabetical

Docentes

Ordenar por:

☒ Nome ☐ Categoria ☐ Área

Lista de Docentes

- Alysson Neves Bessani
- Ana Luísa Respício
- Ana Paula Afonso
- Ana Paula Cláudio
- André O. Falcão
- Antónia Lopes
- António Branco
- António Casimiro
- António Ferreira
- Beatriz Carmo
- Carlos Duarte
- Carlos Lourenço
- Carlos Mão de Ferro
- Carlos Teixeira
- Dimitris Mostrous
- Dulce Domingos
- Emanuel Santos
- Fernando Ramos
- Francisco M Couto
- Francisco Martins
- Graça Gaspar

Exact schemes

Chronological

Browsing by Issue Date

Jump to a point in the index: (Choose year) (Choose month)

Or type in a year:

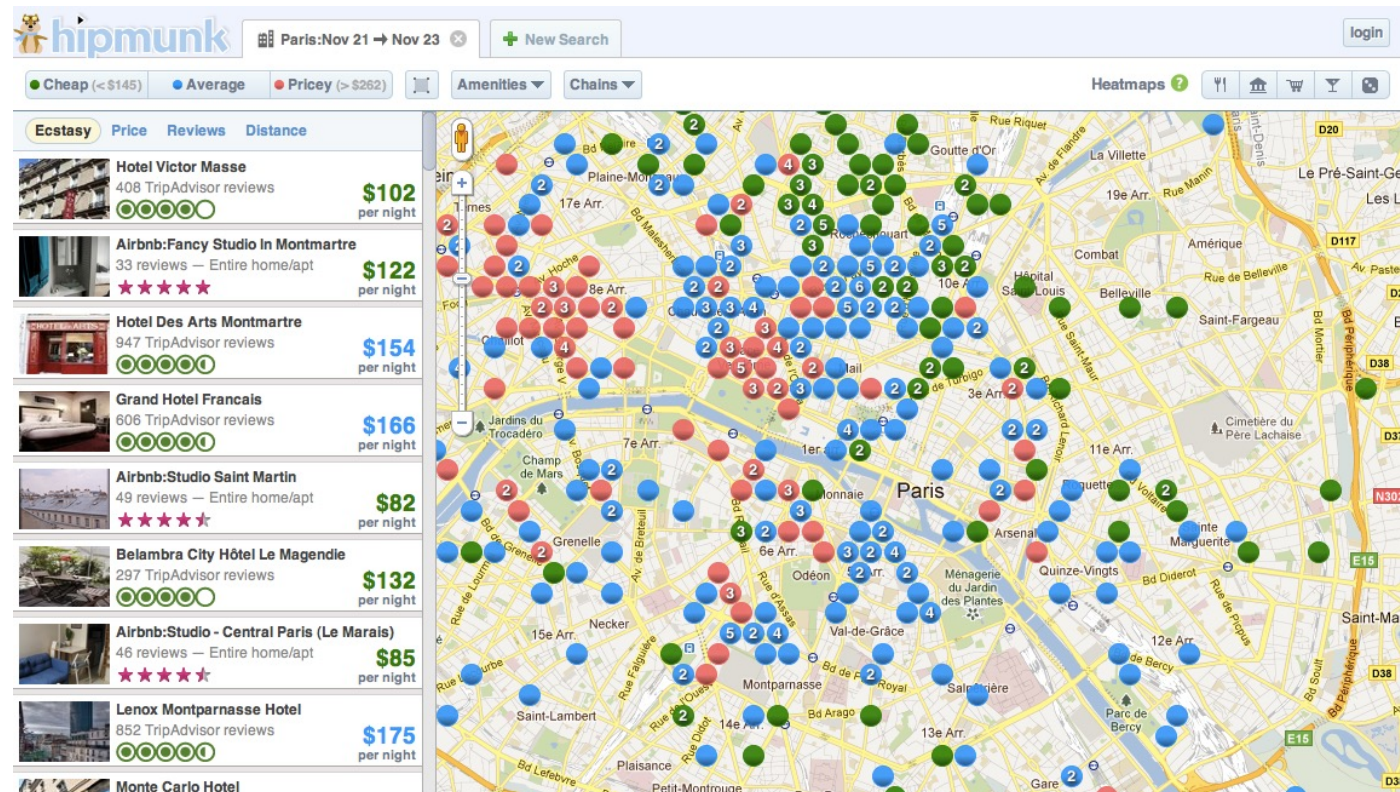
Sort by: In order: Results/Page: Authors/Record:

Showing results 1 to 20 of 1290
[next >](#)

Issue Date	Title	Author(s)
4-Oct-2011	Creating and Assembling Virtual Exhibitions from Existing X3D Models	Cláudio, Ana Paula ; Carmo, Maria Beatriz ; Gomes, Jorge
16-Jul-2011	CINTIL DepBank Handbook: Design options for the representation of grammatical dependencies	Francisco, Costa ; João, Silva ; Sérgio, Castro ; António, Branco
16-Jul-2011	CINTIL TreeBank Handbook: Design options for the representation of syntactic constituency	Sérgio, Castro ; Francisco, Costa ; João, Silva ; António, Branco
6-Jun-2011	ADAPTATION AND LEARNING OF INTELLIGENT AGENTS IN INTERACTIVE ENVIRONMENTS	Policarpo, Daniel Álvaro Fonseca
Apr-2011	CONFIDENTIALITY, INTEGRITY AND NON-REPUDIATION IN SMARTGRIDS	Moniz, Paulo Miguel Santos
17-Jan-2011	ReverX: Reverse Engineering of Protocols	Verissimo, Paulo ; Neves, Nuno ; Antunes, João
2011	CAN CAN ROBÓTICO	Lourenço, Marco António Morais
2011	O MEDIADOR DO EPIDEMIC MARKETPLACE	Ferreira, Hugo Miguel Martins Brito
2011	Realistic Vulnerability Injections in PHP Web Applications	Vieira, Francisco José Marques
2011	SECURITY METRICS TO EVALUATE QUALITY OF PROTECTION	Bruno, Alberto Manuel Giroto

Exact schemes

Geographical



Exact schemes

- A complementary combination of organization schemes is often necessary.

Organization schemes

- Ambiguous
 - Divide information into categories that defy exact definition
 - Difficult to design and maintain. Can be difficult to use.
 - E.g.: Supermarket
 - Why people find them so useful? Because we don't always know what we're looking for!
 - Information seeking is often iterative and interactive!

Organization schemes

- Ambiguous
 - This information-seeking process can involve an element of associative learning
 - Someone other than the user has made an intellectual decision to group items together
 - The success depends upon the quality of the scheme and the careful placement of items within that scheme
 - Rigorous user testing is essential!

Ambiguous schemes

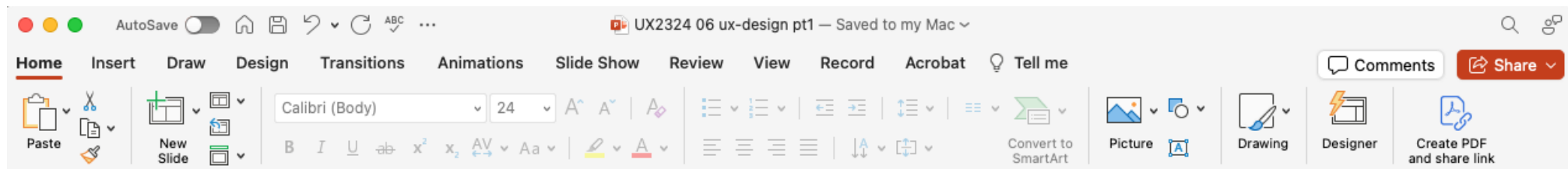
- Topic
 - E.g.: Yellow pages



Shop All Departments	
Books	>
Film, Music & Games	>
Kindle	>
Electronics	>
Computers & Office	>
Home, Garden & Pets	>
Toys, Children & Baby	>
Clothes, Shoes & Watches	>
Hobbies, Sports & Leisure	>
Grocery, Health & Beauty	>
DIY, Tools & Car	>

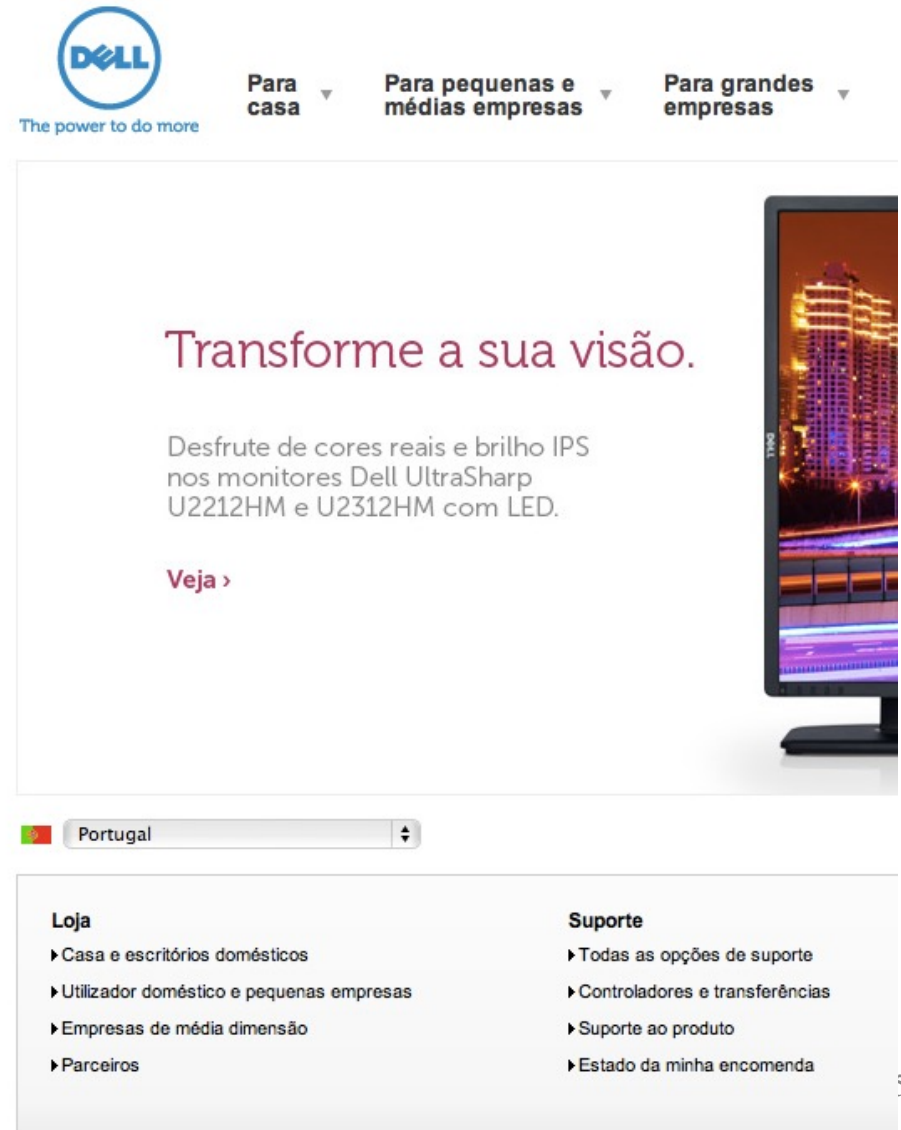
Ambiguous schemes

- Task



Ambiguous schemes

- Audience
 - Can be open or closed



Ambiguous schemes

- Hybrids
 - The power of a pure organization scheme derives from its ability to suggest a simple mental model that users can quickly understand.
 - Sometimes it is “required” to employ hybrid schemes
 - Many web sites have successfully combined topics and tasks on their main page and within their global navigation (e.g.: e-commerce)
 - Schemes combining more categories become hard to understand
 - Unless the design clearly shows the separation between categories

Ambiguous schemes

- Hybrids

Loja

- ▶ Casa e escritórios domésticos
- ▶ Utilizador doméstico e pequenas empresas
- ▶ Empresas de média dimensão
- ▶ Parceiros

Suporte

- ▶ Todas as opções de suporte
- ▶ Controladores e transferências
- ▶ Suporte ao produto
- ▶ Estado da minha encomenda

Informações sobre a empresa

- ▶ Acerca da Dell
- ▶ Sala de imprensa
- ▶ Carreiras

Organization structures

- Movies
 - Linear, experienced frame by frame from beginning to end
 - The plot might be nonlinear!
- Maps
 - Spatial
 - Items are placed according to physical proximity

Organization structures

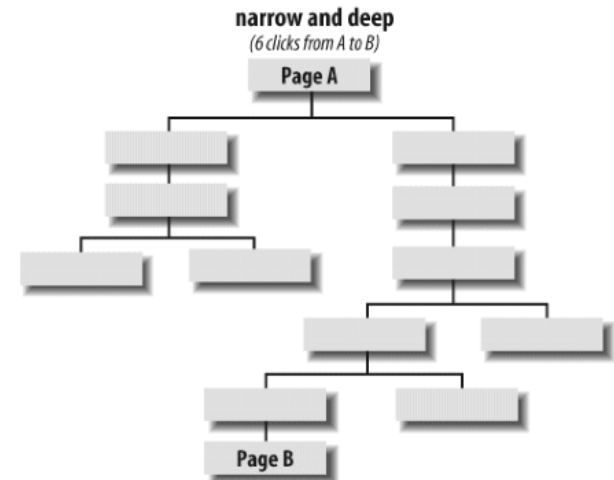
- Hierarchy (or taxonomy)
 - Top-down
 - Foundation of almost all good information architectures
 - Mutually exclusive subdivisions (typically)
 - Parent-child relationships
 - Examples:
 - Family trees
 - Books (chapters, sections, paragraphs, sentences, words, letters)
 - Because it is so pervasive users can easily and quickly understand web sites that use hierarchical organization models

Organization structures

- Designing taxonomies
 - Should be mutually exclusive (not mandatory)
 - Polyhierarchical taxonomies (non exclusive) might be useful when classifying ambiguous information.
 - If the information is available in more than one class then it will be easier to find!
 - But if we take it to far, then the taxonomy loses its meaning!

Organization structures

- Designing taxonomies
 - Balance between breadth and depth
 - If a hierarchy is too narrow and deep, users have to click through an inordinate number of levels to find what they are looking for.
 - Don't make your users navigate more than 2 or 3 levels in the hierarchy

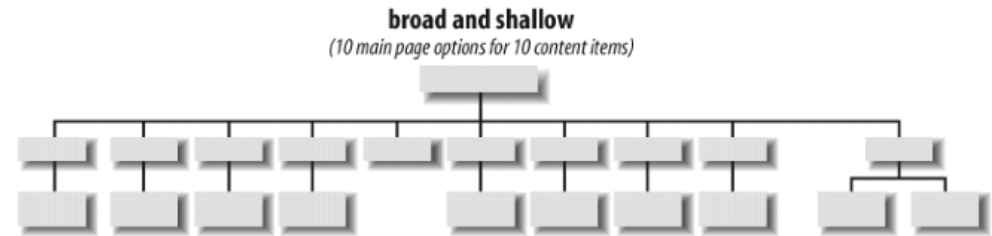


Organization structures

- Designing taxonomies

- Balance between breadth and depth

- If a hierarchy is too broad and shallow users are faced with too many options on the main menu and are unpleasantly surprised by the lack of content once they select an option.
 - Be sensitive to people's visual scanning abilities and to the cognitive limits of the human mind.



Organization structures

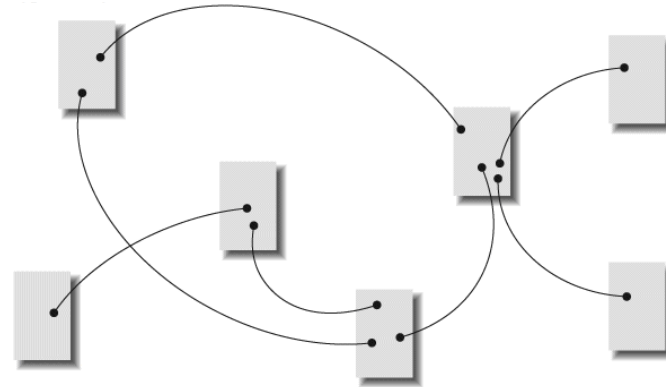
- For new websites that are expected to grow
 - Favor broad and shallow taxonomies over narrow and deep
 - It is less problematic to add items to secondary levels than to the main page!

Organization structures

- Database model
 - Bottom-up
 - Metadata
 - It allows us to apply the structure and power of relational databases to the heterogeneous, unstructured environments of web sites and intranets.
 - By tagging documents and other information objects with controlled vocabulary metadata, we enable powerful searching, browsing, filtering, and dynamic linking.
 - Defining the metadata and relationships can be a complex task, but it is paramount for the success of the information architecture.

Organization structures

- Hypertext
 - Nonlinear structure
 - Nodes and links
 - Provides great flexibility, but presents substantial potential for complexity and user confusion
 - Hypertext links reflect highly personal associations
 - “Lost in hyperspace”
 - Hypertext is rarely a good candidate for the primary organization structure. Rather, it can be used to complement structures based upon the hierarchical or database models.



Organization structures

- Social classification
 - High profile sites have demonstrated the potential to enlist users in content creation and classification
 - When large numbers of people get involved, interesting opportunities arise to transform user behavior and tagging patterns into new organization and navigation systems

Explore / Tags / nature / clusters

Jump to:



[macro](#), [flower](#), [insect](#), [blue](#), [plant](#), [yellow](#), [orange](#), [bee](#), [white](#), [canon](#)

➔ [See more in this cluster...](#)



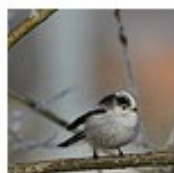
[landscape](#), [sky](#), [flowers](#), [leaves](#), [tree](#), [red](#), [pink](#), [fall](#), [light](#), [cloud](#)

➔ [See more in this cluster...](#)



[green](#), [trees](#), [water](#), [forest](#), [garden](#), [plants](#), [bokeh](#), [rocks](#), [color](#), [verde](#)

➔ [See more in this cluster...](#)



[bird](#), [animal](#), [birds](#)

➔ [See more in this cluster...](#)

These are the *most recent* uploads tagged with **nature**. [See more...](#)

Organization structures

- Social classification

- Issues

- There is no controlled vocabulary, so it might be harder to find information
 - It has been successful only in specific contexts (e.g.: photography)

Organization systems

- When considering organization schemes
 - Exact schemes are best for known-item searching, when users know precisely what they are looking for.
 - Ambiguous schemes are best for browsing and associative learning, when users have a vaguely defined information need.
 - Whenever possible, use both types of schemes.

Organization systems

- When considering organization structures
 - Large web sites and intranets typically require several types of structure.
 - The top-level, umbrella architecture for the site will almost certainly be hierarchical.
 - Keep a lookout for collections of structured, homogeneous information.
 - These potential subsites are excellent candidates for the database model.
 - Less structured, more creative relationships between content items can be handled through author-supplied hypertext or user-contributed tagging.

LABELING SYSTEMS



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Localização

Na Universidade de Lisboa, a Faculdade de Ciências fica na Cidade Universitária, no extremo norte do Jardim do Campo Grande, em Lisboa (ver na área F4 no [mapa da cidade de Lisboa](#)). Pode também ver a nossa localização no [mapa interativo da Google](#):



Labeling

Goal: to efficiently communicate information

Labeling

- It's something natural
 - Names!
- It's the easiest way to present a website's organization and navigation systems
- Being so natural makes it easy to take it for granted (which often results in poor labels)

Labeling

- When we talk to another person, we rely on constant user feedback to help us hone the way we get our message across.
- When we converse with users through the website we design the feedback isn't quite immediate, if it exists at all.
- Information architects must design labels that speak the same language as a site's users.

Example

Main

[Get Rates & Reservations](#)

[Find a U-Haul Location](#)

[College Connection](#)

[Hot Deals](#)

Products & Services

[Trucks & Trailers](#)

[Self-Storage](#)

[Boxes & Moving Supplies](#)

[Vehicle Transportation](#)

[Dollies & Furniture Pads](#)

[Propane](#)

[RoofPouch](#)

[Hitches](#)

[Carpet Cleaners](#)

SuperGraphics

[Pictorial Tribute to North America](#)

Corporate

[Customer Service](#)

[Public Relations](#)

[Dealer Opportunities](#)

[Truck Sales](#)

[Real Estate](#)

[Corporate Move](#)

[Company History](#)

[Missing or Abandoned Equipment](#)

[Career Opportunities](#)

Buy Online

[The U-Haul Store](#)

Example

- Main
 - The main page? Are the links related to the main page?
- SuperGraphics
 - What does this means?
 - What can we expect to find there and in the only link in the group?
- Corporate
 - Is this for visiting users or for employees (or corporate users)?
 - What does it mean “Corporate Move”?
 - Shouldn’t “Truck Sales” be inside “Products & Services”?
- Buy Online
 - Only one link again!
 - Why isn’t the store in the “Products & Services” section?

Problems

- The labels aren't representative and don't differentiate
 - What does it mean “Corporate Move”?
 - What distinguishes “Products & Services” from the store?
- The labels are jargony, not user-centric
 - “College connection” or “SuperGraphics”
- The labels don't make a good impression

Categories

- Textual
 - Contextual links
 - Headings
 - Navigation system choices
 - Index terms
- Graphical
 - Iconic

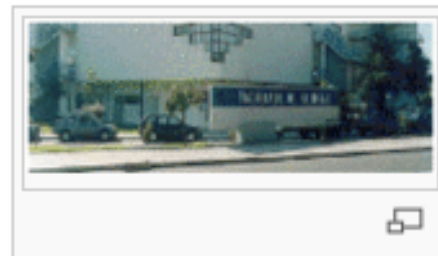
Contextual links

- Hyperlinks
- Generally not developed systematically, but in an ad hoc manner when the author makes a connection between his text and something else
- They are more personal, which makes it easier for them to have different meanings to different people
- The meaning is very dependent on the surrounding context (the surrounding text is fundamental to their understanding)

Informática na Universidade de Lisboa, Faculdade de Ciências

O Departamento de Informática da Universidade de Lisboa [🔗](#), Faculdade de Ciências [🔗](#) está reconhecidamente entre as melhores escolas de ensino superior do país na área da engenharia informática.

Admitimos mais de 150 novos alunos de licenciatura por ano nas licenciaturas em Engenharia Informática e em Tecnologias de Informação e Comunicação. Contamos com mais de 1 000 alunos de doutoramento, mestrado e licenciatura.



Promovemos iniciativas interdisciplinares que resultam das nossas parcerias com as mais reputadas empresas de tecnologias de informação e das ligações estreitas com outras escolas e departamentos, tanto dentro da universidade como a nível internacional.

Estamos na Cidade Universitária [🔗](#), um dos melhores lugares para fazer um percurso académico, com os seus espaços amplos, as suas instalações desportivas e localização ímpar, perto de tudo em Lisboa.

Contextual links

- How to make sure they are representative
 - What kind of information will the user expect to be taken to?
 - Ask yourself
 - Ask others

Headings

- Describe the chunk of information that follows
- Establish a hierarchy within a text

Palestras e Seminários

Seminários do Departamento

Os [Seminários do Departamento de Informática](#) são um fórum de discussão e difusão de matérias relevantes para a investigação, ensino e divulgação da Informática.

São públicos e abertos à participação de todos os interessados.

Provas Académicas

[Ver lista de Provas Académicas](#)

Tipos de provas académicas públicas:

- Provas de Agregação
- Provas de Doutoramento ([provas com registo em filme](#))
- Provas de Qualificação para Doutoramento
- Provas de Mestrado

As teses estão disponíveis no [Repositório do Departamento de Informática](#)

Seminários da Pós-Graduação

[Ver lista de Seminários](#)

São palestras dos alunos de doutoramento e têm normalmente lugar lectivo.

Headings

- Graphic design can help establish hierarchies thus assisting the information architect's job
- When used in processes
 - Using numbers communicates progression and using verbs frames the labels as actions

Labels within navigation systems

- The small number of options demands consistent application of labels
 - Repeatedly exposed throughout the site



Labels within navigation systems

- Common variants (by “category”)
 - Main, Main Page, Home – Início, Entrada
 - Search, Find, Browse – Pesquisar, Procurar
 - Contact, Contact Us – Contatos, Contate-nos

Index terms

- Can be used to describe any type of content
- Support more precise searching than simply searching the full text of content
 - Someone has assessed the content's meaning and described it using index terms
 - Can serve as metadata source for browsable lists

Index terms

Explore the BBC

Full A-Z

Click on a letter

0 - 9	A	B	C	D	E	F	G	H	I	J	K	L	
M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

-  A Hard Day's Night
-  A Question of Sport
- Abkhazia (Territory Profile)
- About the BBC
-  Absolute Power
- Accessibility
-  According to Bex
-  Across the Line
-  Adam Walton
- Adult Learning
-  Advice (Radio 1)
- Afghanistan profile
- Africa News
- Ajaria (Region Profile)
- Alba homepage
- Albania (Country Profile)
- Albanian (World Service)
- Algeria profile
-  All Things Considered

Iconic

A picture is worth a thousand words
but which thousand?

Iconic



Iconic



Designing Labels

- Guidelines
 - Remember *content, users* and *context*
 - Narrow scope whenever possible
 - You might have to use different label categories in different areas of the site if it is too broad
 - Develop consistent labelling systems

Consistency

- Style
 - Punctuation and case
- Presentation
 - Fonts, font sizes, colors, whitespace, grouping
- Syntax
 - Verbs, names, questions
- Granularity
 - Chinese, Italian, Sushi, McDonald's
- Comprehensiveness
 - Pants, ties, shoes
- Audience
 - Stroke, tummy ache

Designing Labels

- Where to start
 - Labels currently on your site and competitor's sites
 - What works (and what doesn't)
 - Create a labeling table and identify patterns (they will inform your choice of labels)
 - Controlled vocabularies and thesauri
 - Useful for indexing content
 - Important for specific audiences (with their own specific vocabulary)

Designing Labels

- Creating new labeling systems
 - Content analysis
 - Content authors
 - User advocates and subject matter experts
 - Directly from users
 - Card sorting
 - Brainstorming
 - Indirectly from users
 - Search-log analysis (search analytics)
 - Tag analysis

Designing Labels

- Creating new labeling systems
 - Sort the list (alphabetically) and remove duplicates
 - Review for consistency
 - Look for gaps
 - Keep the scope narrow and focused
 - Be prepared to review your labels!

NAVIGATION SYSTEMS

Navigation



Start by

Ensuring good organization

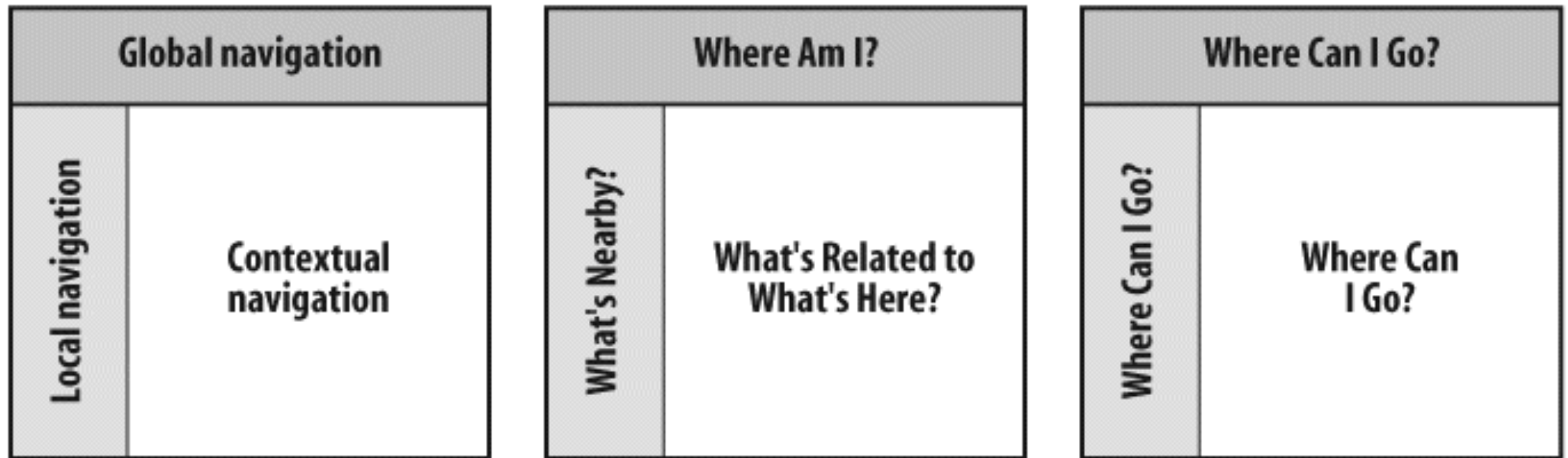


Navigation adds



Types of navigation systems

- Global, local, contextual



Types of navigation systems

- Sitemaps, indexes, guides, ...

Sitemap	Index	Guide
Category1 <u>SubCat1, Subcat2, Subcat3</u>	A <u> </u> <u> </u>	Step1 <u> </u> <u> </u>
Category2 <u>SubCat1, Subcat2, Subcat3</u>	B <u> </u> <u> </u>	Step2 <u> </u> <u> </u>
Category3 <u>SubCat1, Subcat2, Subcat3</u>	C <u> </u> <u> </u>	Step3 <u> </u> <u> </u>

Even before you start to build your site, there are navigation tools provided by the ...

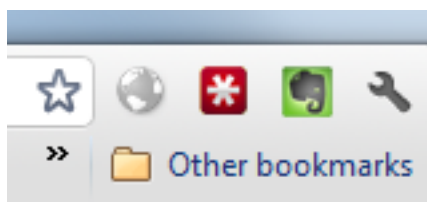
Browser

- Open URL 

- Back/Forward

- History

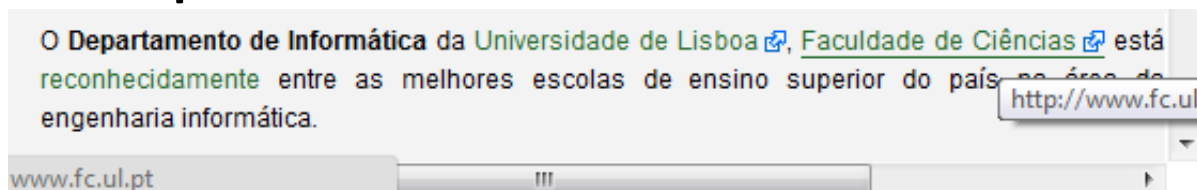
- Favorites



- Link color

expressing a **model** or **concept** of **information** used in

- Prospective view



History

Today - Thursday

10:51 

10:51 

10:49 

10:49 

10:48 

10:47 

10:46 

10:42 

10:40 

10:40 

Avoid

O Departamento de Informática da Universidade de Lisboa [↗](#), Faculdade de Ciências [↗](#) está reconhecidamente entre as melhores escolas de ensino superior do país na área da engenharia informática.

- Modifying visited/unvisited link colors
- Killing the back button

Web navigation context

- No natural landmarks
- Non-linear navigation
- Users can bypass the home page

Web navigation context

- User should always know which site they're in
- Present the structure of the information hierarchy in a clear and consistent manner, and indicate the user's current location



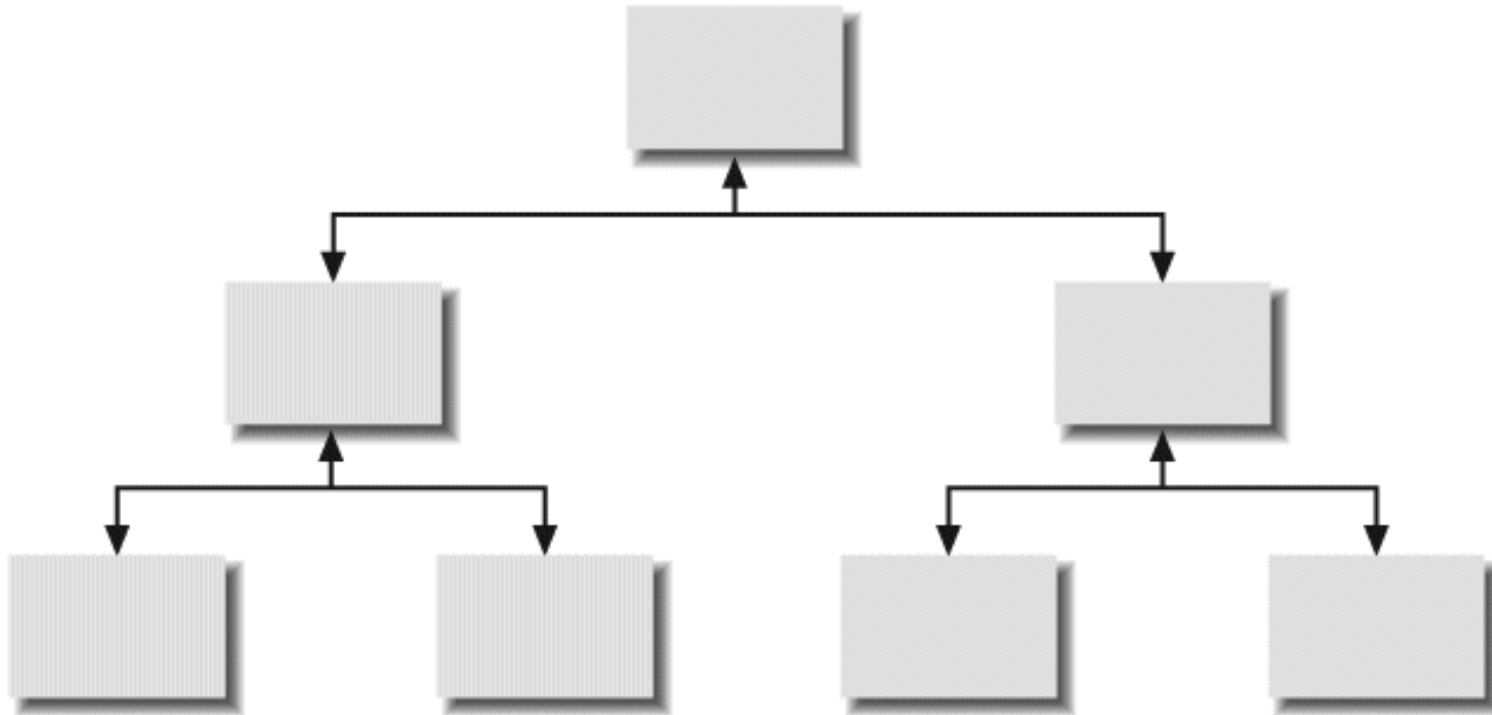
Web navigation context

- Navigation stress test (Instone 1997)
 - Ignore the home page and jump directly into the middle of the site
 - For each random page, can you figure out where you are in relation to the rest of the site? What major section are you in? What is the parent page?
 - Can you tell where the page will lead you next? Are the links descriptive enough to give you a clue what each is about? Are the links different enough to help you choose one over another, depending on what you want to do?

Information structure is a great place to start designing your navigation, but ...

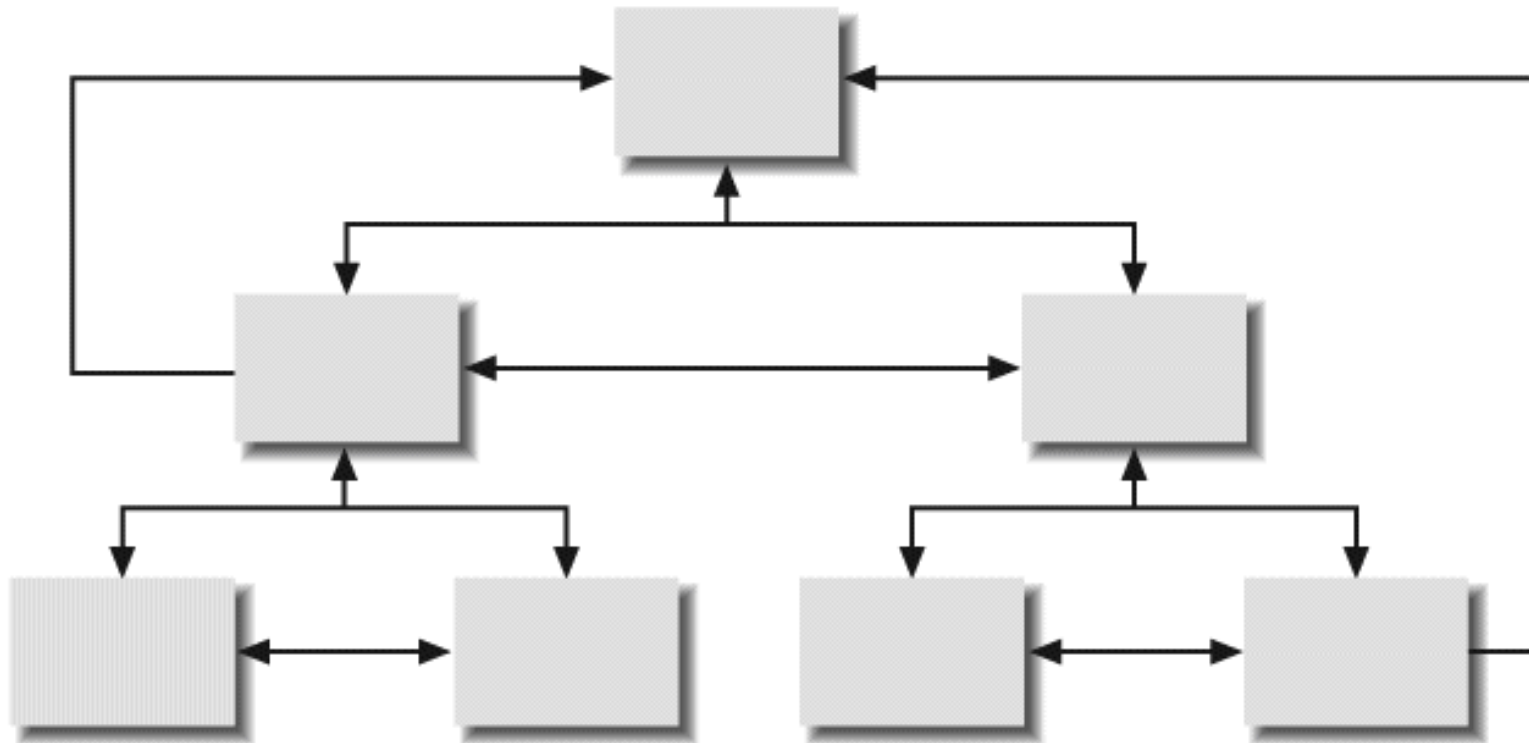
Flexibility

- Hierarchies limit flexibility



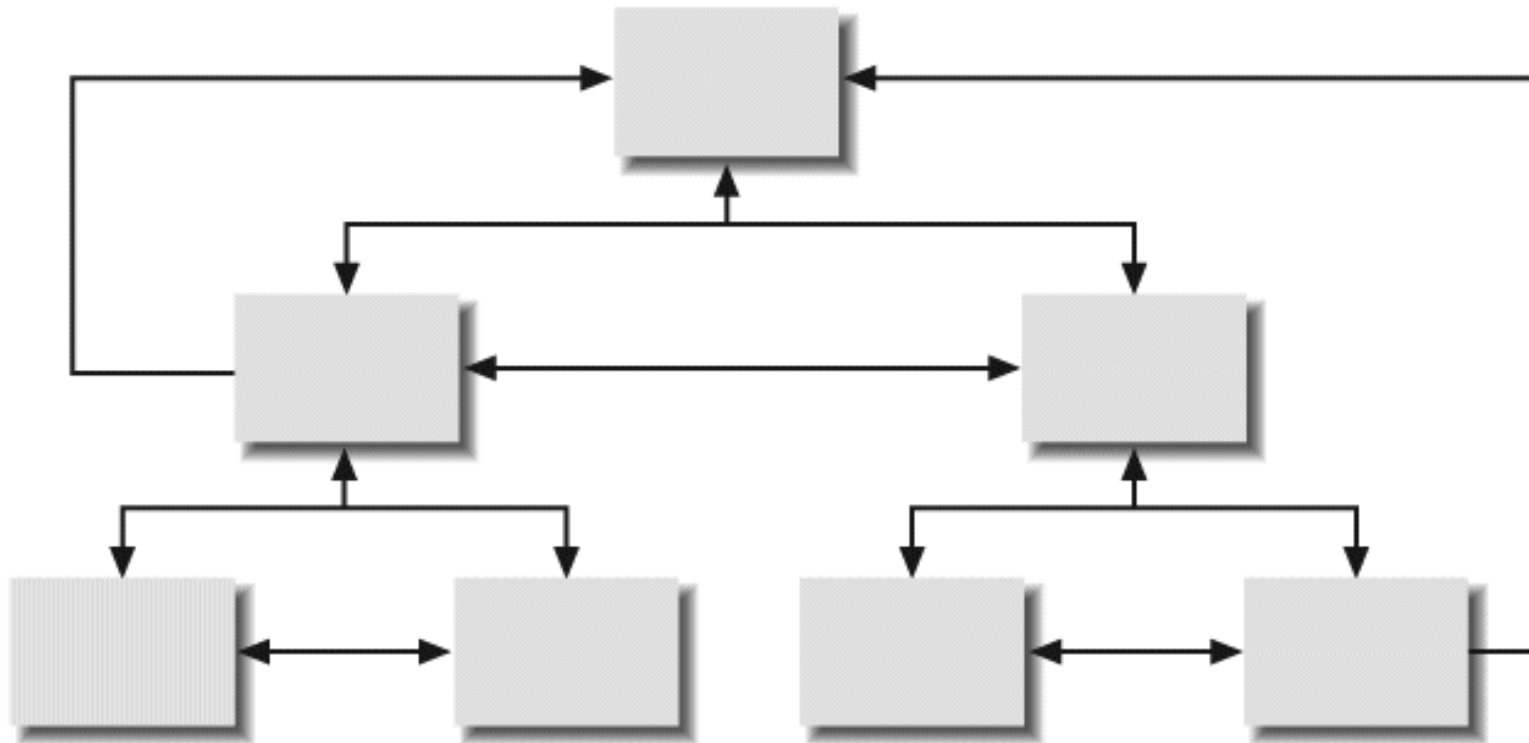
Hypertext

- Flexibility



Hypertext

- Confusing



EMBEDDED NAVIGATION SYSTEMS

Global navigation	
Local navigation	Contextual navigation

Where Am I?	
What's Nearby?	What's Related to What's Here?

Where Can I Go?	
Where Can I Go?	Where Can I Go?

Global navigation system

- Available on every page throughout a site
 - Direct access to key areas and functions
- Often implemented in the form of a navigation bar at the top of each page
- Most global navigation bars provide a link to the home page
 - The main page is sometimes the sole exception to the omnipresence of the global navigation bar! In other cases, the main page presents a variety of navigation options, which makes it difficult to understand which one is the global!
- Many include search

Global navigation system

- Being an omnipresent element, its design demands special care
 - Test, test, test
- For complex sites (thousands of pages) it is important that it is compact

Feedback zum neuen Design | Produkte kaufen | Empfehlen | Alle Microsoft-Seiten | Deutschland | **Microsoft**

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Produkte | Office | Windows | Kaufen | Downloads | Leistungen für | Sicherheit | Support | Über Microsoft

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Jetzt Hotmail nutzen

Hotmail

Wechseln auch Sie jetzt zu Hotmail >

Ich möchte

- den neuesten Browser nutzen
- Windows Internet Explorer 9
 - Internet Explorer 9 Testsieger der Computer Bild >
 - Herunterladen >
- kostenlos E-Mail nutzen
- meine Freunde im Video-Chat treffen

Produce und Downloads

Aktuell | Windows | Office | Xbox | Windows Phone | Windows Live | Hardware | Zune



Wichtige Sicherheitsupdates

Laden Sie die Updates vom 11. Oktober herunter

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Windows 7 Home Premium

Windows 7 - mehr Unterhaltung auf dem PC.

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Übertreffen Sie sich selbst - holen Sie sich Ihr Office-Paket.

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Windows Live Tipps und Tricks >

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Aktuelle Downloads

DirectX >

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Microsoft Office Compatibility Pack für Dateiformate von Word, Excel und PowerPoint >

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Aktuelle News

Steve Jobs (1955-2011) >

Cloud 4 Society Award >

Microsoft Security Intelligence Report: No Patch - No Fun >

Hamburger Studenten entdecken den Nerd in sich >

Aktuelle Pressemitteilungen >



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WÄHREND SIE FOTOS TEILEN

Windows Live Messenger 2011

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Descubra o Cloud Power

Faça muito mais, com muito menos.

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Windows Azure, a plataforma da cloud para desenvolver e fornecer aplicações

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Trabalhe na cloud pública ou privada com as suas ferramentas actuais

» Saiba mais

Steve Jobs (1995-2011) Comunicado de Steve Ballmer, CEO da Microsoft

Destaques

Recentes lançamentos

Utilizar o computador

Para empresas

Para profissionais de TI

Para programadores

Soluções

CRM Online
Experimente Grátis durante 30 dias. Até ao final do ano pode também subscrever em condições especiais

Eleve o seu negócio
Conheça uma forma completamente nova de trabalhar.

Abertura Microsoft
Como criar valor utilizando Open Source com a Microsoft.

Experimente o Novo Windows Intune
Assegure a gestão e a segurança dos seus PCs de forma simplificada. Experimente gratuitamente durante 30 dias.

Cloud Power
Que outra solução pode dar mais recursos?

Segurança
Faça a actualização de segurança

Top Downloads

- Internet Explorer 9
- Windows 7
- Windows Live Messenger

Pesquisas Frequentes

- Modelos
- Encontre a Aplicação ou Solução que necessita!
- Messenger
- Windows Live Messenger

Principais Destinos

- Serviço de Apoio a Clientes
- Download Center
- MSDN para Programadores
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[Free download](#)
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Latest news

- Microsoft expands data platform with SQL Server 2012; new investments
- LA Fitness, Volkswagen choose Lync to unify communications
- New research shows browsers are front line of defense: Are you safe?

See the latest news from Microsoft

Popular searches

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- Fun
- Downloads & Articles
- For work
- Security
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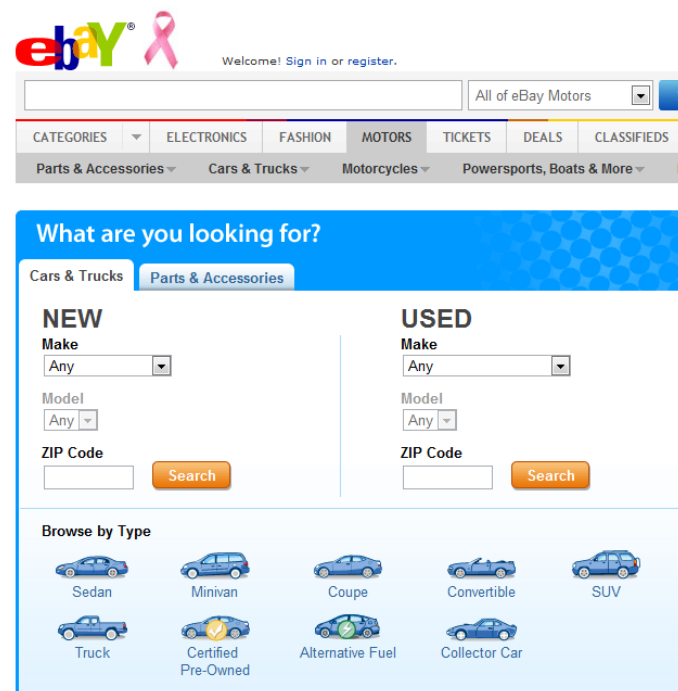
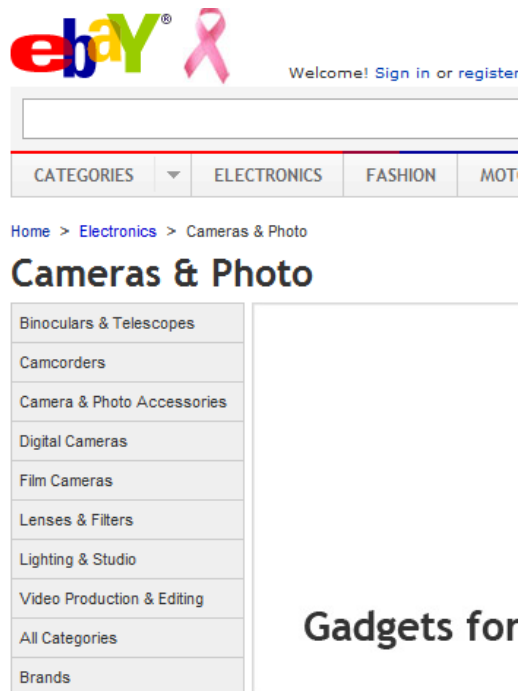
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Local navigation system

- Complement global navigation systems



Local navigation system

- Promote the organization of related contents
- Might result from different teams' developments

Contextual navigation system

Some relationships
don't fit neatly into
the structured
categories of global
and local navigation

The screenshot shows the Amazon.com homepage for a user named CARLOS DUARTE. The navigation bar includes a search bar, a dropdown menu for 'Books', and links to 'Advanced Search', 'Browse Subjects', 'New Releases', 'Best Sellers', and 'The New York'. A 'Fall Reading' banner is prominently displayed. The main content area is divided into several sections: 'Browse Books' with a sidebar for 'Amazon Kindle' and 'Best Books'; 'Books' with a section for '2011 National Book Award Finalists'; 'Fall Reading' with a section for 'What's hot in Books this fall'; and 'Recommended for You' with a section for 'Look Inside!'. The 'Look Inside!' section features five books with their covers and titles: 'Understanding Shutter Speed: Creative...', 'Measuring the User Experience...', 'The Design of Everyday Things', 'Interaction Design: Beyond...', and 'Research Methods in Human-Computer...'. Each book has a 'Fix this recommendation' link.

amazon.com Hello, CARLOS DUARTE. We have [recommendations](#) for you. (Not CARLOS?)
CARLOS's Amazon.com | [Today's Deals](#) | [Gifts & Wish Lists](#) | [Gift Cards](#)

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Fall Reading [Shop now](#)

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Books
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2011 National Book Award Finalists
21 National Book Award [finalists](#) were announced on October 12. Judges selected finalists in four young people's literature. Among the finalists are [Téa Obreht's](#) novel *The Tiger's Wife* and *Malcol* who passed away just days before the book was published. Winners will be announced on Nov

- [Fiction finalists](#)
- [Nonfiction finalists](#)
- [Poetry finalists](#)
- [Young People's Literature finalists](#)

[See all 2011 National Book Award finalists](#)

Fall Reading
What's hot in Books this fall

Lee Child **48% off**
The Affair
Fall Blockbusters

Haruki Murakami **47% off**
1Q84
Literature & Fiction

Inheritance **50% off**
Inheritance
Teens

Karen **50% off**
Reckless
Mystery

[See the full list](#)

Recommended for You

LOOK INSIDE!
Understanding Shutter Speed: Creative...
Bryan Peterson
Paperback
\$25.95 **\$16.21**
[Fix this recommendation](#)

LOOK INSIDE!
Measuring the User Experience...
Tom Tullis, Thomas Tullis, William Albert
Paperback
\$49.95 **\$38.59**
[Fix this recommendation](#)

LOOK INSIDE!
The Design of Everyday Things
Donald A. Norman
Paperback
\$16.95 **\$11.53**
[Fix this recommendation](#)

LOOK INSIDE!
Interaction Design: Beyond...
Helen Sharp, Yvonne Rogers, Jenny Preece
Paperback
\$16.95 **\$11.53**
[Fix this recommendation](#)

LOOK INSIDE!
Research Methods in Human-Computer...
Jonathan Lazar, Jinjuan Heidi Feng, ...
Paperback
\$48.98
[Fix this recommendation](#)

[See more recommendations](#)

Contextual navigation system

- Associative learning
- Resulting from editorial decisions, not architectural (typically)
- Should not be critical for navigation
 - Users can easily ignore them
- When used inline special care should be given to the link's text

Navigation bars

- Images vs. text



Navigation bars

- Dynamic

The image displays two versions of the eBay website's navigation bar, illustrating dynamic content. The top version shows a standard navigation bar with a search bar, category dropdown, and links to 'My eBay', 'Sell', 'Community', and 'Customer Support'. The bottom version shows a more detailed view of the navigation bar, including a list of categories and a 'Popular on eBay' section.

Top Screenshot:

- Navigation Bar:** Includes the eBay logo, a search bar, and a dropdown menu for 'All Categories'. Links for 'My eBay', 'Sell', 'Community', and 'Customer Support' are present.
- Content Area:** Features a large banner for 'eBay Stickers and StubHub have teamed up!' and a sidebar with category links.

Bottom Screenshot:

- Navigation Bar:** Similar to the top version, but with a more detailed category list on the left.
- Content Area:** Includes a 'Welcome to eBay' section, a 'Shop safely on eBay' section, and a 'Sign in' section. A 'Popular on eBay' section at the bottom shows four iPhone 4s devices.

SUPPLEMENTAL NAVIGATION SYSTEMS

- Because embedded systems will fail for a significant percentage of users and tasks

Sitemap	Index	Guide
Category1 <u>SubCat1, Subcat2, Subcat3</u>	A <u> </u> <u> </u>	Step1 <u> </u> <u> </u>
Category2 <u>SubCat1, Subcat2, Subcat3</u>	B <u> </u> <u> </u>	Step2 <u> </u> <u> </u>
Category3 <u>SubCat1, Subcat2, Subcat3</u>	C <u> </u> <u> </u>	Step3 <u> </u> <u> </u>

Sitemaps

Table Of Contents

Submission	3
Preface	7
Executive Summary	9
Action Plan Priority List	17
Introduction	19
Statement of Goals	21
Land Use Element	29
Part A. Land Use Maps and Studies	31
Part B. Community Character and Design	67
Transportation Element	81
Public Facilities Element	91
Sensitive Areas Element	99
Mineral Resources Element	113
Historic Resources Element	117
Appendices	
A. Population and Trends	121
B. Land Use Data	125
C. Previous Studies	139
D. "Cottage Housing" Examples	147
E. Street Design Standards	151
F. Excerpt from the State Development Capacity Task Force ...	159
G. Chesapeake Bay Critical Area	161

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---	--	--	---	---

Can't find it? Try the **A to Z**



Sitemaps

- Broad view of the content in the website
- Facilitates access to segmented portions of the content
- Useful from a search engine optimization perspective
- More useful for larger sites
- Most natural for sites that are hierarchically organized

Site Indexes

- Works well for users that know what they are looking for
- Granularity
 - Web pages? Concepts?
 - What the users are looking for!
 - Know what your users need

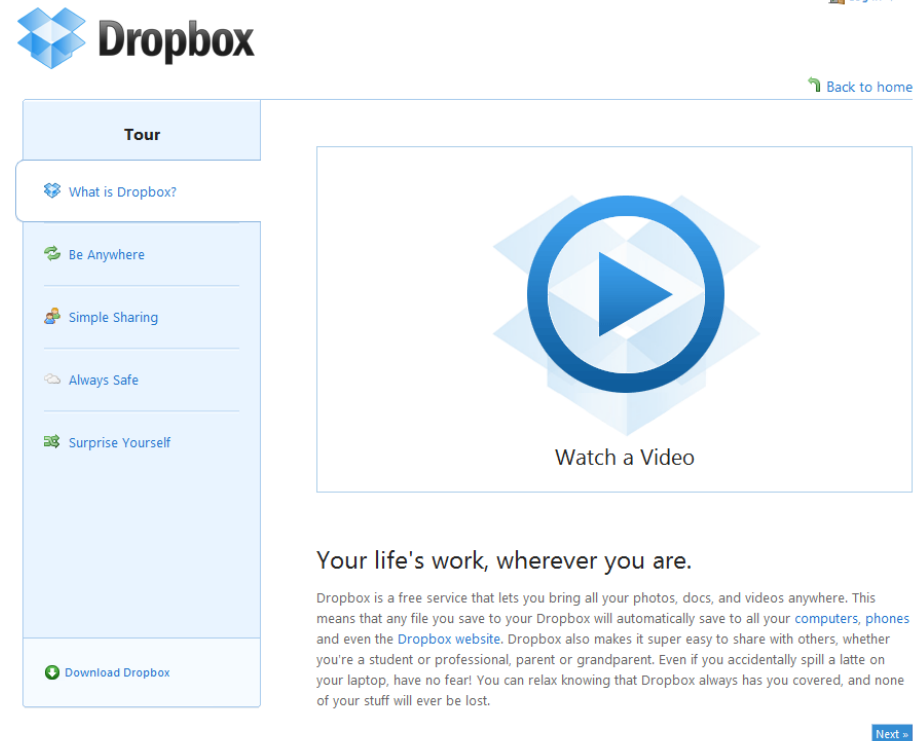


Site Indexes

- Creating a site index
 - Manually
 - For smaller sites
 - Entries are selected by the author or an editor
 - Automatically
 - Controlled vocabulary indexing

Guides

- Guided tours
- Tutorials
- Micro-portals
- Typically feature linear navigation
 - Help new users
 - Complement with hypertext



Guides

- Should be short
- At any point, the user should be able to exit the guide
- Navigation (previous, home, next) should be in the same spot on every page of the guide
- Should be designed to answer questions
- Screenshots should be crisp, clear and optimized, with enlarged details of key features
- If the guide includes more than a few pages, it may need its own table of contents

Search

- It also is supplemental navigation
- Puts the user in control
- Needs to deal with ambiguity of the language

SEARCH SYSTEMS

Does your site need search?

- Do not assume that a search engine alone will satisfy all user's information needs.
 - Some users are natural browsers, preferring to forego search!

Does your site need search?

Consider:

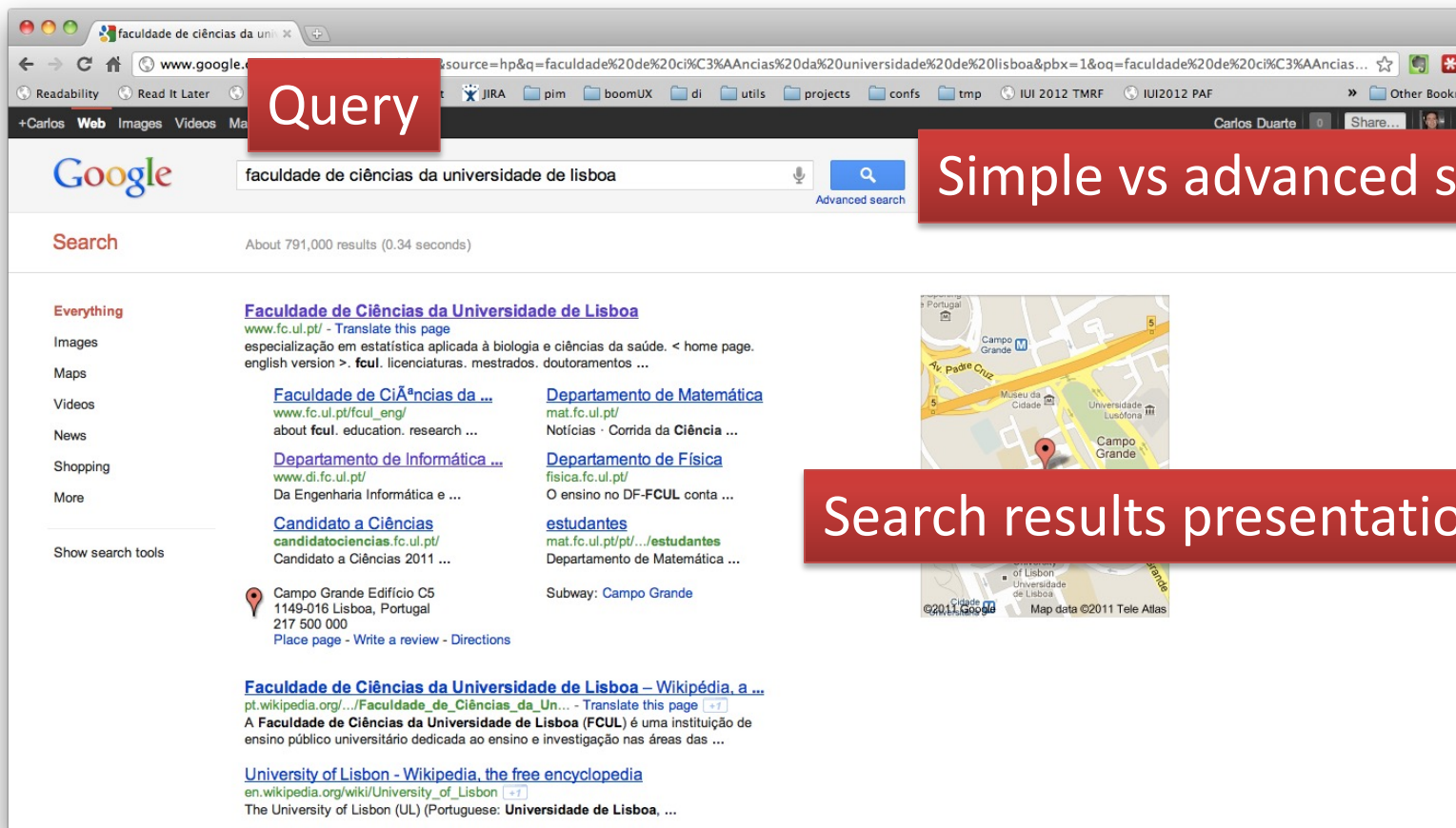
1. Does your site have enough content?
2. Will investing in search systems divert resources from more useful navigation systems?
3. Do you have the time and know-how to optimize your site's search system?
4. Are there better alternatives?
5. Will your site's users bother with search?

Does your site need search?

Issues:

1. Search helps when you have too much information to browse
2. Search helps fragmented sites
3. Search helps to get to know your users
4. Users expect it to be there
5. Search can tame dynamism

Search system anatomy



The screenshot shows a Google search interface. The search bar contains the text "faculdade de ciências da universidade de lisboa". A red box labeled "Query" points to the search bar. The search results are displayed below the search bar, showing a list of results for "Faculdade de Ciências da Universidade de Lisboa". A red box labeled "Simple vs advanced search" points to the search bar. The search results include a map of the university campus, a list of departments (e.g., Departamento de Matemática, Departamento de Física), and a list of search results (e.g., Candidato a Ciências, Candidato a Ciências 2011 ...). A red box labeled "Search results presentation" points to the search results.

Query

Simple vs advanced search

Search results presentation

WHAT TO SEARCH

Indexing

- Everything
 - Not the best solution all the times
- Search zones
 - Pockets of more homogeneous content
 - Allows users to focus their searches
- Content components
 - Author vs. copyright statement
 - Not all information has the same value

Search zones

Search Results for "xt3"

► Search within Small & Medium Business

Narrow your selection:

Product Category

- Parts & Upgrades (5)
- Computer Components (4)
- Memory (3)
- Laptop/Notebook Accessories (1)
- Laptop/Notebook Power (1)

See Results from:

Help & Support ?

- Drivers and Downloads (86)
- Troubleshooting (1)

News & Articles ?

- News (1)

Community Discussions ?

- Blogs (7)
- Forums (4)
- TechCenter Wiki (55)

Find Results In:

- Home & Home Office
- Small & Medium Business
- Large Business
- State & Local Government
- Federal Government
- K-12 Institutions
- Higher Education Institutions
- Healthcare

Products

Help & Support

Solutions & Services

News & Articles

Community Discussions

Page View: [Grid](#) | [List](#)

Showing 4 Results

Sort by: [Relevance](#)

RECOMMENDED



New Workplace, New Latitude E-Series

More and more organizations are going virtual and in the next 10 years, Gen-Y will make up more than 48% of the workforce. This group is tech savvy, information hungry and they know what they want. Before we started planning the new Latitude E-series... (Read more on our blog Direct2Dell)



Dell

1 GB Dell Certified Replacement Memory Module for Dell Latitude XT3 Laptop

Usually Ships Within 24 Hours
Manufacturer Part# SNP3D9HMC/1G
Dell Part# A5256348

- Technology: DDR3 SDRAM
- Memory Speed: 1333 MHz (PC3-10600)
- Data Integrity Check: Non-ECC

Starting Price.....**\$17.99**

► More Details

➔ Add to Cart



Dell

4 GB Dell Certified Replacement Memory Module for Dell Latitude XT3 Laptop

Usually Ships Within 24 Hours
Manufacturer Part# SNPX830DC/4G
Dell Part# A5256350

- Technology: DDR3 SDRAM
- Memory Speed: 1333 MHz (PC3-10600)

Starting Price.....**\$44.99**

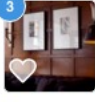
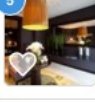
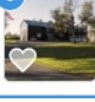
Search zones

 Oslo, Norway: Tue Nov 24 - Wed Nov 25 1 room, 1 person [add this search to a trip](#) ▼

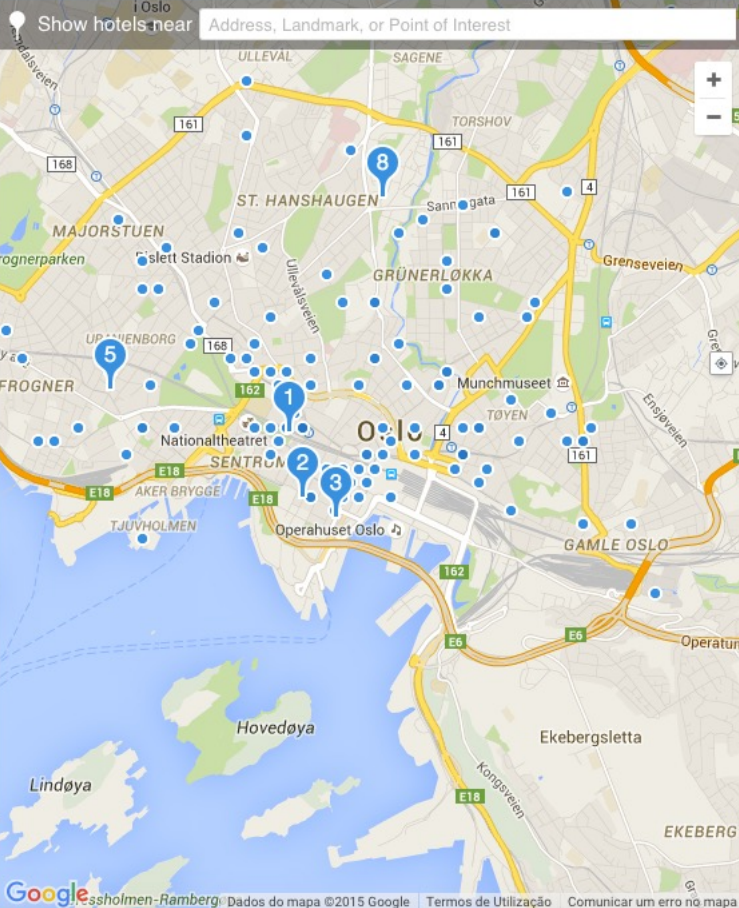
Filters ▶ [view more](#) **Price:** \$ \$ \$ \$ \$ **Stars:** ☆☆☆☆☆ & up **Reviews:** & up

Sort: Ecstasy Price Stars Reviews Savings   Heatmaps ▼  My Saved Hotels

216 Results Available

-  **Grand Hotel**   ~~186 €~~ **160 €** 
-  **First Hotel Grims Grenka**   **151 €**  4 Rooms left!!
-  **Clarion Collection Hotel ...**   **112 €**  791 reviews
-  **Soria Moria Hotell**    ~~134 €~~ **113 €**  503 reviews 3 Rooms left!!!
-  **Frogner House Apartmen...**   **77 €** 
-  **Lysebu Hotel**    **113 €**  576 reviews

My Saved Hotels ▲



Google  Essholmen-Ramberg Dados do mapa ©2015 Google Termos de Utilização Comunicar um erro no mapa

Search zones

- Subsets of a web site that have been indexed separately from the rest of the site's content
- By eliminating content that is irrelevant to his needs, the user should retrieve fewer, more relevant, results... but can be more complex!
- Content structure might/should help determining search zones

Content components

ALWAYS REMEMBER YOU'RE ONE OF A KIND. WE WILL.

FIND US ON   **GOYOU** 

TOPIC

American History

Follow American History 

WEDNESDAY, OCT 19, 2011 1:00 AM WEST

The store that killed small business

A new book explores how A&P transformed the retail industry and set the stage for our chain-dominated world

BY BARBARA SPINDEL, BARNES & NOBLE  72



TOPICS: AMERICAN HISTORY, NONFICTION, BOOKS

This article appears courtesy of the [Barnes & Noble Review](#).

"Big Business Now Sweeps Retail Trade," declared the New York Times in 1928. "Huge Corporations, Serving the Nation Through Country-Wide Chains, Are Displacing the Neighborhood Store." While some version of that headline wouldn't be out of place in a present-day report about Walmart, the chain that provoked the most anxiety almost a century ago was the Great Atlantic & Pacific Tea Co., which in 1920 began a 43-year run as the largest retailer in the world, transforming the retail industry in ways that made Walmart possible.

In his eye-opening book ["The Great A&P and the Struggle for Small Business in America,"](#) Marc Levinson covers the history of A&P, the industry it

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GOYOU 

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POST JOIN ALL POSTS

- Occupy Oakland in the 'Town: This revolution will be t
PAMELA MAYS McDONALD
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STEPHANIE RUSSELL-KRAFT
- It all comes down to this, America: Don't be Cruel
RAMONAS VOICES

Currently In Salon


Pat Buchanan declares defeat


Barbie gets a tattoo makeover


Hillary's legacy rests on fixing tainted pipeline approval process


Record number of deportations still not enough for anti-immigration zealots


In praise of the

Recommendations


Barbie gets a tattoo makeover
109 people recommend this.


This is really the best the GOP can do?
187 people recommend this.


Voting, not OWS, will save America
417 people recommend this.


Pat Buchanan declares defeat
45 people recommend this.

196

Content components

- Is it used in searches?
 - Not by the majority of users...
 - Advanced searches
- Content components aren't useful only for enabling more precise searches; they can also make the format of search results much more meaningful

PRESENTING SEARCH RESULTS

Main issues

- Which content components to display for each result?
- How to list or group the results?

Which components to display

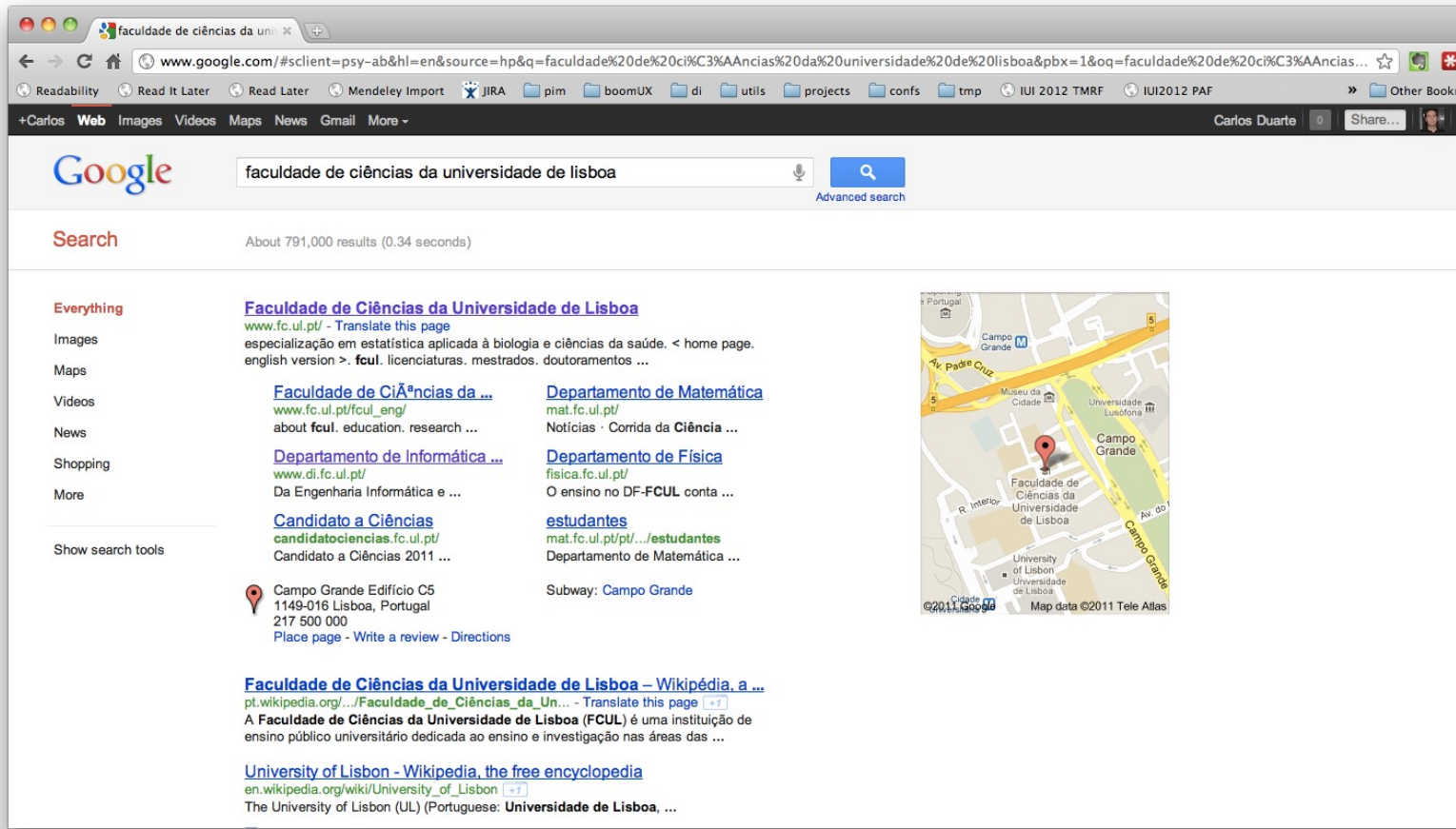
Guideline:

- Display less information to users who know what they are looking for, and more information to users who aren't sure what they want

Consider also:

- How much information to display per result is also a function of how large a typical result set is

Which components to display



The screenshot shows a Google search interface with the query "faculdade de ciências da universidade de lisboa". The search results are categorized under "Everything" and include links to the official website, various departments (Mathematics, Physics, Informatics), and Wikipedia entries. A map on the right shows the location of the Faculty of Sciences at Campo Grande in Lisbon. A blue arrow points from the word "fold" to the bottom of the search results, indicating a scroll action.

Search About 791,000 results (0.34 seconds)

Everything

- Images
- Maps
- Videos
- News
- Shopping
- More
- Show search tools

Faculdade de Ciências da Universidade de Lisboa
www.fc.ul.pt/ - Translate this page
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Faculdade de Ciências da Universidade de Lisboa – Wikipédia, a ...
pt.wikipedia.org/wiki/Faculdade_de_Ci%C3%Aancias_da_Universidade_de_Lisboa - Translate this page
A Faculdade de Ciências da Universidade de Lisboa (FCUL) é uma instituição de ensino público universitário dedicada ao ensino e investigação nas áreas das ...

University of Lisbon - Wikipedia, the free encyclopedia
en.wikipedia.org/wiki/University_of_Lisbon
The University of Lisbon (UL) (Portuguese: **Universidade de Lisboa**, ...

Map
Map data ©2011 Tele Atlas

fold

Which components to display

- Which components are available?
 - Are there labels? Are the documents structured?
- What are the information needs of the users?
 - If a user is looking for a phone number in a directory, it makes sense to show that information in the search result itself

Show the query terms in the context of the document itself

[Tips for better searching](#)

Results for *hedge* in E*TRADE Financial

E*TRADE Monthly Market Commentary

.S. Treasury strength, particularly in shorter maturities. Source: Municipal Market Advisors The chart above is illustrative of lack of buying interest in the longer portion of the curve despite clear value. Other than retail investors, the vast majority of potential municipal buyers view long-dated bonds less favorably due to a higher concentration of bond option risk and the lack of **hedging** options

Help Center: Trade on margin Trade on margin

positions or to specific pre-defined options strategies (e.g. Spreads, Covered Calls/Puts, Straddles, etc.). Typically, portfolio margin treatment provides greater leverage than strategy-based treatment for a portfolio of **hedged** positions. This means that investors who manage position risk by **hedging** positions with related securities will be able to invest in more market share with less capital commitment

OPEN AN ACCOUNT

. Request a Brochure GLOBAL TRADING FEATURES Trade online in 6 global markets including Canada, France, Germany, Hong Kong, Japan, and the UK. 1 Manage exchange-rate risk and **hedge** U.S. dollar exposure. 2 Seize international growth opportunities

How many documents to display

- Suggestions:
 - Let the user know the total number of retrieved documents
 - Provide a results navigation system to help the user move through the results

1-10 of 2.780.000 results

1 2 3 4 5 6 7 8 9 10 Next >

Listing results

- Sorted
 - Chronologically
 - Alphabetically
 - Title, Author, Department, ...
 - Price
 - Especially helpful for users who are looking to decide or take an action

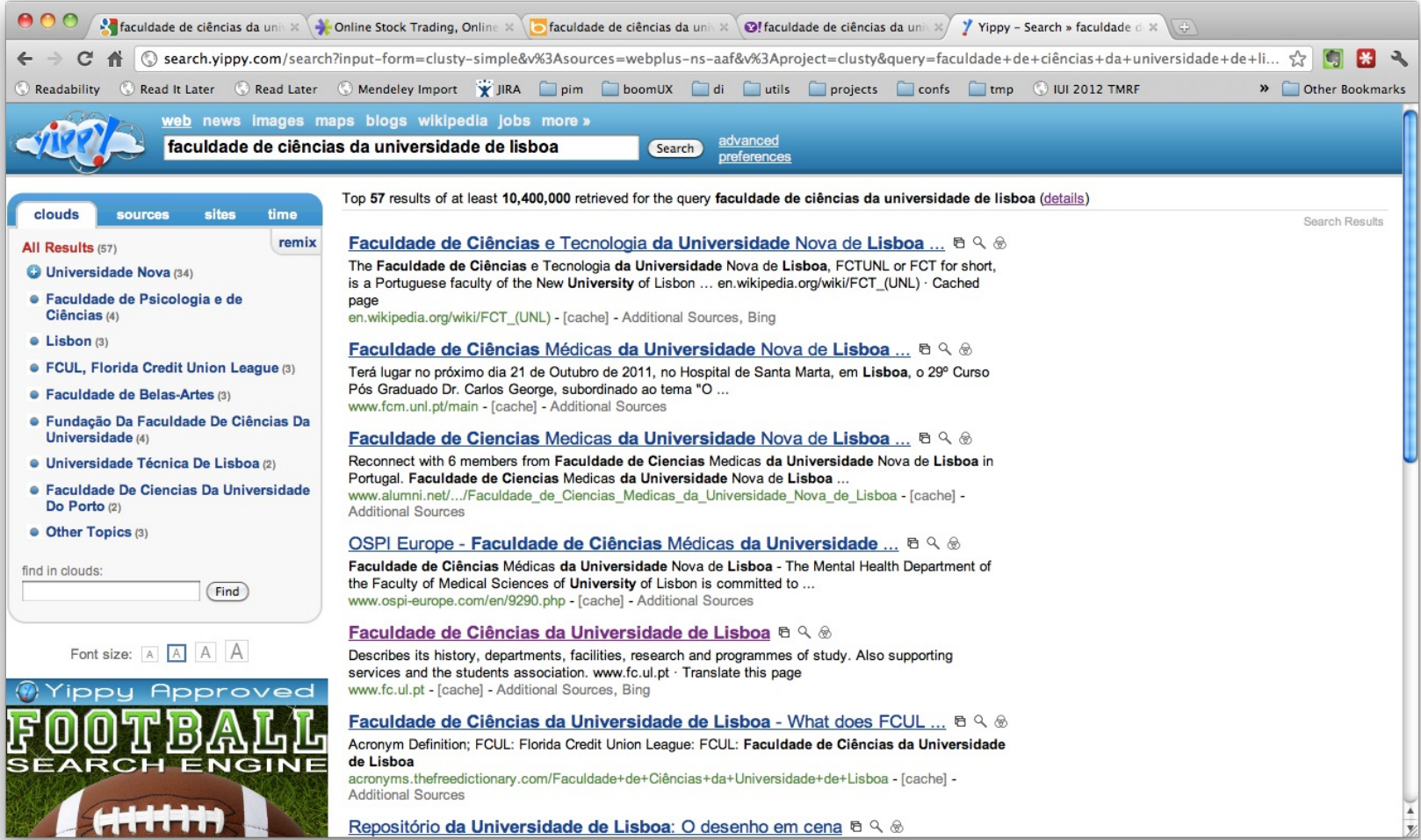
Listing results

- Ranked
 - Relevance (is relative!)
 - Popularity
 - Users
 - Pay-for-placement
 - More useful when there is a need to understand information of learn something

Grouping results

- Criteria
 - Document type, creation/modification date
 - Easier to obtain, but less useful
 - Manually applied metadata: topic, audience, product family, ...
 - More useful, but can be prohibitively expensive

Grouping results



The screenshot shows a web browser window with multiple tabs. The active tab is Yippy, displaying search results for the query "faculdade de ciências da universidade de lisboa". The browser's address bar shows the Yippy search URL. The Yippy interface includes a search bar, navigation links (web, news, images, maps, blogs, wikipedia, jobs, more), and a sidebar with filters (clouds, sources, sites, time). The main content area displays a list of search results, each with a title, description, and source. The results are grouped by relevance, with the top result being "Faculdade de Ciências e Tecnologia da Universidade Nova de Lisboa".

Top 57 results of at least 10,400,000 retrieved for the query **faculdade de ciências da universidade de lisboa** (details)

Faculdade de Ciências e Tecnologia da Universidade Nova de Lisboa ...
The Faculdade de Ciências e Tecnologia da Universidade Nova de Lisboa, FCTUNL or FCT for short, is a Portuguese faculty of the New University of Lisbon ... en.wikipedia.org/wiki/FCT_(UNL) · Cached page
en.wikipedia.org/wiki/FCT_(UNL) - [cache] - Additional Sources, Bing

Faculdade de Ciências Médicas da Universidade Nova de Lisboa ...
Terá lugar no próximo dia 21 de Outubro de 2011, no Hospital de Santa Marta, em Lisboa, o 29º Curso Pós Graduado Dr. Carlos George, subordinado ao tema "O ...
www.fcm.unl.pt/main - [cache] - Additional Sources

Faculdade de Ciências Médicas da Universidade Nova de Lisboa ...
Reconnect with 6 members from Faculdade de Ciências Médicas da Universidade Nova de Lisboa in Portugal. Faculdade de Ciências Médicas da Universidade Nova de Lisboa ...
www.alumni.net/.../Faculdade_de_Ciencias_Medicas_da_Universidade_Nova_de_Lisboa - [cache] - Additional Sources

OSPI Europe - Faculdade de Ciências Médicas da Universidade ...
Faculdade de Ciências Médicas da Universidade Nova de Lisboa - The Mental Health Department of the Faculty of Medical Sciences of University of Lisbon is committed to ...
www.ospi-europe.com/en/9290.php - [cache] - Additional Sources

Faculdade de Ciências da Universidade de Lisboa
Describes its history, departments, facilities, research and programmes of study. Also supporting services and the students association. www.fc.ul.pt · Translate this page
www.fc.ul.pt - [cache] - Additional Sources, Bing

Faculdade de Ciências da Universidade de Lisboa - What does FCUL ...
Acronym Definition; FCUL: Florida Credit Union League: FCUL: Faculdade de Ciências da Universidade de Lisboa
acronyms.thefreedictionary.com/Faculdade+de+Ciências+da+Universidade+de+Lisboa - [cache] - Additional Sources

Repositório da Universidade de Lisboa: O desenho em cena

Next week

- T class
 - UX design
- TP class
 - Student presentations
 - UX design